Worker Bargaining Power and Market Tightness: Theory and Evidence *

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Job Market Paper

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Abstract

What determines worker bargaining power? This paper argues that bargaining power is an equilibrium outcome of labour market conditions: when alternative employers are easier to find, a worker's threat to walk away from a negotiation becomes credible, and the worker's bargaining power rises. I formalise this in a non-cooperative bargaining model in which workers and firms keep searching while they negotiate. In equilibrium, bargaining power is strictly increasing in labour market tightness. The model delivers an empirical counterpart: bargaining power is the pass-through of firm productivity into wages, which should therefore rise with tightness. Using French matched employer-employee data and measures of local labour market tightness, I find that a one-euro increase in the hourly marginal product of labour raises the hourly wage by 3.3 cents at average tightness and by 3.7 cents one standard deviation above it, a 13% increase in pass-through.

Keywords: Worker bargaining power, labour market tightness, labour share of revenue, search frictions

JEL Codes: C78, D83, J21, J31, J42, L13

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1 Introduction

When a worker and a firm enter an employment relationship, they create value. When this value increases, workers capture only part of the gain: a 10% increase in value added per worker raises wages by 0.5 to 1.5% (Card et al., 2018). The share captured by workers reflects their bargaining power, which varies systematically across countries, firms, worker types, and over time (Cahuc et al., 2006; Stansbury and Summers, 2020; Mengano, 2022; Bell et al., 2024). Labour market tightness, the number of vacancies per job seeker, is a natural candidate to explain this variation: workers should be able to capture more of the gains when they are more difficult to replace.

While the intuition is simple, formalising it has proved challenging. Once a worker and a firm meet to negotiate a wage, it is not obvious how opportunities elsewhere should affect the wage they negotiate. Hall and Milgrom (2008) argue that outside options should matter little, because the credible threat is to delay agreement rather than return to search. Others allow labour market competition to affect wages through other channels such as on-the-job search or threat points, while keeping the bargaining weight fixed (Cahuc et al., 2006; Jarosch et al., 2024). An open question is whether market conditions can alter bargaining power itself.

This paper shows that they can. I develop a theory of wage bargaining that combines three features. First, workers and firms bargain by *alternating offers*, taking turns to propose a wage. Second, there is *continued search*: while negotiating, both sides may be paired with new potential partners before the counteroffer is made. Third, these new opportunities do not mechanically end an ongoing negotiation. Agents make an *endogenous partner choice*, deciding whether to continue with their current partner or pursue the new one. If they choose a new partner, the previous negotiation breaks down before a counteroffer can be made. The key innovation is this third feature: by making partner choice endogenous, the model determines in equilibrium whether and how an outside opportunity affects the ongoing negotiation.

Together, these three features generate an endogenous risk that a negotiation breaks down after a rejected offer. This risk depends naturally on how easily each side finds an alternative partner, but also on bargaining roles. In equilibrium, each side obtains a larger value when acting as proposer than when acting as responder. An agent is therefore more reluctant to abandon a negotiation in which they are about to propose than one in which they would respond. Consequently, the probability that a negotiation survives differs depending on who rejected the previous offer: the worker or the firm.

The probability that a negotiation survives, in turn, determines the value of rejecting

an offer. In this alternating offers framework, the value of an agreement for each party is ultimately determined by the *value of rejecting*: whoever makes an offer must guarantee the other party a value at least as large as that of rejecting and continuing to negotiate. Importantly, obtaining a higher value does not necessarily imply greater bargaining power. What matters is the value of agreement *relative to the value of the outside option*, which is the value of continuing the search by abandoning the negotiation. This difference is each party's surplus, and the share of total surplus they obtain represents their bargaining power.

Labour market tightness affects these two values through opposing channels. Consider a worker who receives a wage offer from a firm. If the worker rejects, they become the proposer in the following round, provided the negotiation continues. First, a tighter labour market makes it less likely that the firm finds another worker and abandons the negotiation. The worker is therefore more likely to have the opportunity to voice their counteroffer. This raises the value of continuing to negotiate and, consequently, the value of the agreement the firm must offer.

However, a tighter labour market also increases the value of the worker's outside option, their unemployment value. An unemployed worker benefits from finding a new firm, regardless of who gets to propose. By contrast, a worker who has just rejected an offer already expects to propose in the ongoing negotiation. When they encounter a new firm that would propose, they choose to remain in the ongoing negotiation, where they have the more valuable role. This choice is optimal, but comes at the cost of foregoing an opportunity that would have been valuable if unemployed. As such opportunities become more frequent in a tighter labour market, the value of unemployment rises more than the value of continuing to negotiate through this channel. Since surplus is the difference between the value of agreement and the value of unemployment, this puts downward pressure on the worker's bargaining power.

Despite these opposing forces, the model delivers a sharp result. In the symmetric benchmark, where workers and firms are equally likely to make the first offer, worker bargaining power is strictly increasing in labour market tightness. I then allow the probability of making the first offer to differ across the two sides. The positive relationship continues to hold as long as market tightness sufficiently changes the relative ease with which workers and firms find alternative partners. The relationship remains positive across standard pairing technologies and a wide range of empirically plausible parameter values. It can reverse only when the firm's probability of finding another worker is almost unresponsive to tightness: in that case, the worker's opportunity cost of delay rises while the reduction in breakdown risk is negligible.

This benchmark isolates how bargaining power varies across labour markets that have

different degrees of market tightness, but abstracts from an important source of variation within them: matches differ in the total value they create. I therefore extend the model to allow for firm heterogeneity. A worker negotiating with a more productive firm requires a more attractive alternative before it becomes worthwhile to abandon the current negotiation. As the productivity of the current firm rises, a smaller fraction of the outside opportunities encountered by the worker therefore become credible alternatives. The extension implies that more productive firms pay higher wages, but also identifies a force that can reduce workers' share of match surplus as firm productivity rises. This mechanism can also weaken the effect of labour market tightness on bargaining power, and may even reverse it among highly productive firms.

The theory provides a direct way to take these predictions to the data. In the model, worker bargaining power is equal to the pass-through of productivity into wages: the share of a marginal increase in productivity that is reflected in the worker's wage. The model therefore predicts that pass-through should be higher in tighter labour markets. It further predicts that pass-through should decline with firm productivity, and that its sensitivity to labour market tightness should weaken as firm productivity rises.

I test these predictions using French matched employer–employee data linked to measures of local labour market tightness at the occupation-by-location-by-year level. I estimate industry-specific production functions to construct each firm's hourly marginal revenue product of labour. At the worker level, I relate hourly wages to the marginal revenue product of the worker's firm, local labour market tightness, and their interaction. The interaction is the key empirical object: it measures whether the pass-through of marginal revenue product into wages is higher in tighter labour markets, as predicted by the model.

Because local labour market tightness is jointly determined with wages, I instrument it using the leave-one-out average tightness of the same occupation in other locations. The instrument exploits occupation-wide shifts in labour demand that are common across locations, while excluding tightness in the worker's own local labour market. The identifying variation therefore comes from changes in the demand for an occupation outside the worker's own location. I additionally instrument the firm's current marginal revenue product using its lag, addressing potential endogeneity and measurement error in the productivity measure.

The central prediction is strongly supported by the data. The pass-through of the firm's marginal revenue product of labour into wages increases with labour market tightness. The interaction between marginal revenue product and local tightness is positive and robust across specifications. In the preferred specification with firm fixed effects, a one-euro increase in marginal product of labour raises hourly wages by 3.3 cents at average standardized labour market tightness. A one-standard-deviation increase in tightness raises this pass-through to

3.7 cents, an increase of approximately 13%.

I next examine whether productivity pass-through varies across firms of different productivity. I rank firms by their revenue productivity within industries and regions and estimate pass-through across productivity deciles separately by sector. Revenue productivity determines the firm's position in the distribution, while the marginal revenue product of labour remains the variable whose pass-through into wages I estimate.

The results are consistent with the mechanism of the model. However, the shape of this relationship varies substantially across sectors. In particular, pass-through is relatively flat across the productivity distribution in sectors with slacker labour markets, consistent with the numerical illustration. Taken together, these findings suggest that the role of outside opportunities in wage bargaining depends both on labour market conditions and on the productivity of the firm. They provide further evidence that worker bargaining power is not a fixed parameter, but responds to the environment in which negotiations take place.

Related literature. This paper builds on bargaining theory and the literature studying wage bargaining in frictional labour markets. The seminal paper by Hall and Milgrom (2008) emphasises the cost of delaying agreement, while Rubinstein and Wolinsky (1985) allow search to continue during bargaining. I preserve both forces but make the decision to abandon an ongoing negotiation for a newly available partner endogenous. In contrast to these models, my model delivers bargaining weights that respond positively to the availability of outside options.

The model is closely related to Cahuc et al. (2006), where on-the-job search and outside offers generate competition between employers. Jarosch et al. (2024) show that workers cannot use other vacancies at their current employer as outside options, so larger employers effectively remove more alternatives from the worker's choice set. My paper complements these models by endogenising the bargaining weight which is an exogenous parameter in these models: it is Nash bargaining weight in Jarosch et al. (2024), while in the framework of Cahuc et al. (2006), it is related to the time each side takes to formulate a counteroffer.

Closest to my central prediction, Mangin and Sedláček (2018) and Autor et al. (2023) also imply that tighter labour markets increase workers' share of productivity, through direct firm competition and lower monopsony markdowns, respectively. My mechanism instead arises from two-sided search during bargaining, in which both workers and firms can encounter alternative partners. A further distinction is distributional: my model predicts that bargaining power declines with firm productivity, a pattern I also find in the data.

The empirical analysis relates to two further literatures. First, a large literature documents incomplete rent sharing between firms and workers (Card et al., 2018), while recent

work documents substantial variation in worker bargaining power and rent sharing over time (Stansbury and Summers, 2020; Mengano, 2022). My mechanism provides a potential explanation for these trends. Córdoba et al. (2023) show that the efficient bargaining weight under CES technology can vary with labour market tightness and can contribute to movements in the labour share. I instead derive the bargaining weight from a non-cooperative bargaining game and causally test its implications using worker–firm data. Second, recent empirical work shows that tighter labour markets raise wages, particularly toward the bottom of the wage distribution (Börschlein et al., 2024). Rather than estimating the effect of tightness on wages directly, I ask whether tightness changes the pass-through of firm productivity into wages, providing a direct test of whether the division of surplus varies with labour market conditions.

The remainder of the paper is organised as follows. Section 2 presents the benchmark bargaining model, characterises equilibrium bargaining power, and establishes its relationship with labour market tightness. Section 3 extends the framework to heterogeneous firms and examines how bargaining power varies with firm productivity. Section 4 describes the data and the construction of the main variables. Section 5 presents the empirical strategy, while Section 6 reports the results. Section 7 concludes.

2 Model

I develop a partial equilibrium model of a frictional labour market in which workers and firms bargain over wages while continuing to search for alternative partners.

2.1 Environment

The economy is populated by workers and firms. At any point in time, a mass N_u of workers are unemployed and searching for jobs, while a mass N_v of vacancies are available at firms, each with one position to fill. Labour market tightness is defined as $\theta = \frac{N_v}{N_u}$, the number of vacancies per unemployed worker.

Time is discrete and infinite. Agents are risk neutral and discount future payoffs by a common factor $\delta \in (0, 1)$. An employment relationship produces a period output p, of which the worker receives a wage w and the firm retains $p - w$. At the end of each period, the relationship is destroyed with exogenous separation probability $s \in (0, 1)$, after which the worker returns to unemployment and the firm reopens its vacancy. While unemployed, workers receive a period utility b, whereas firms incur a period cost c to maintain an unfilled vacancy. I assume that employment generates strictly positive joint surplus relative to

continued search, $p - b + c > 0$.

Let V_i^0 denote the value of searching, measured at the start of a period, with $i \in \{w, f\}$, where w denotes a worker and f a firm. It depends on equilibrium behaviour and is deferred to Section 2.5. Given V_i^0, the value of an employment relationship at wage w, measured at the start of a production period, satisfies

$$\begin{aligned} V_w(w) &= w + \delta\big[(1-s)V_w(w) + sV_w^0\big], \\ V_f(w) &= p - w + \delta\big[(1-s)V_f(w) + sV_f^0\big]. \end{aligned} \tag{1}$$

Workers and firms are paired through a frictional process. Each period, an unemployed worker is paired with a firm with probability $\lambda_w(\theta)$, while a firm with an open vacancy is paired with a worker with probability $\lambda_f(\theta)$, where both probabilities lie in $(0, 1)$. Higher market tightness increases the probability that a worker is paired with a firm and decreases the probability that a firm is paired with a worker:

$$\lambda_w'(\theta) > 0, \qquad \lambda_f'(\theta) < 0. \tag{2}$$

Pairings are one to one, implying that $\lambda_w(\theta) = \theta\lambda_f(\theta)$. Pairing draws are independent over time, and each agent can be paired with at most one new partner per period.

I consider a stationary environment in which the masses of unemployed workers and vacancies remain constant over time. Workers and firms that leave the search market following an employment agreement are replaced by new entrants.

Throughout the analysis, I take N_u and N_v, and hence labour market tightness θ, as given. This allows me to isolate how tightness affects the division of surplus between workers and firms. It also reflects the empirical setting, where different labour markets exhibit different levels of tightness. These differences may arise from productivity, labour supply, product market demand, or other factors. Rather than modelling their determinants, I study how bargaining outcomes vary with tightness.

2.2 Timing

Each period consists of a pairing stage followed by a bargaining stage. In the pairing stage, new worker and firm pairs are formed according to the pairing probabilities $\lambda_w(\theta)$ and $\lambda_f(\theta)$. For each new pairing, Nature selects the initial proposer, assigning the role to the firm with probability μ and to the worker with probability $1 - \mu$. In the baseline model, I set $\mu = 1/2$, so that workers and firms are equally likely to make the initial offer.

In the bargaining stage, a worker and a firm negotiate over the wage through the al-

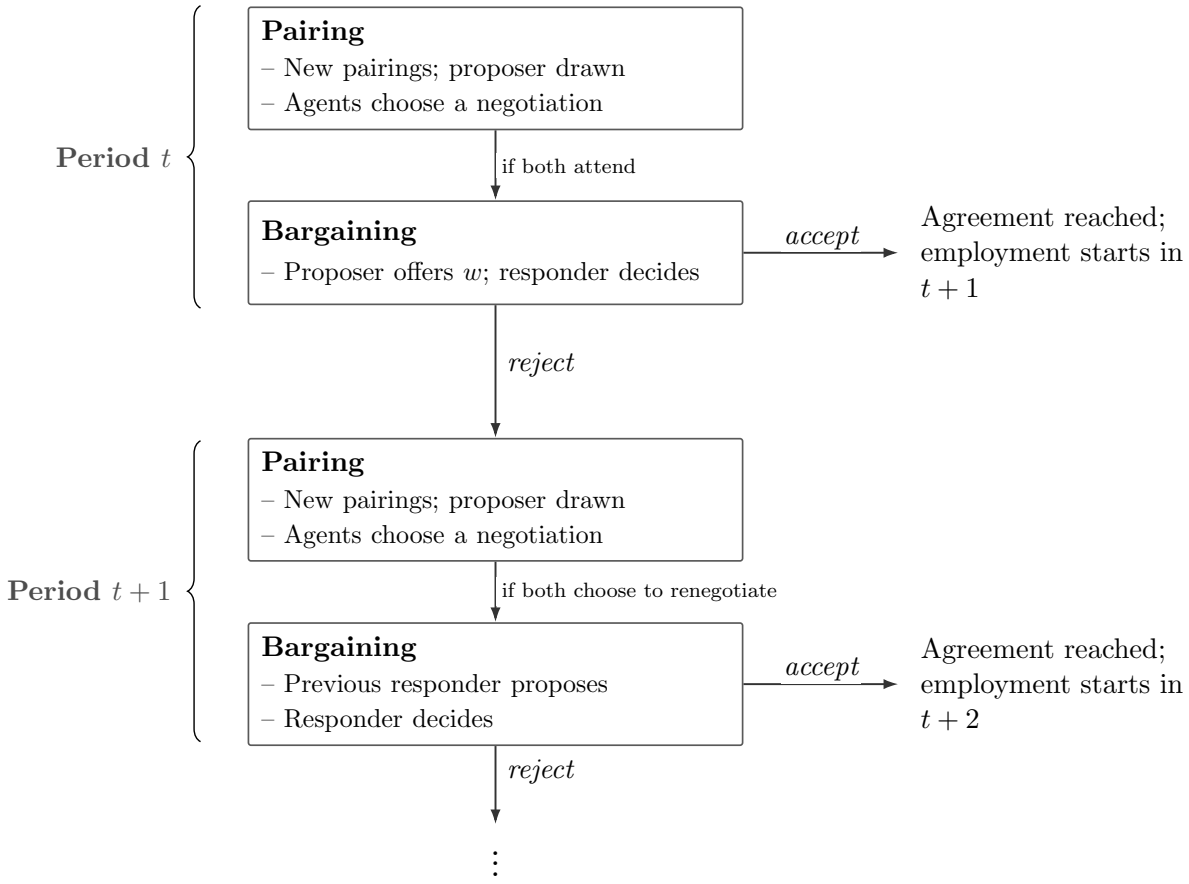


Figure 1: Timeline of a worker–firm negotiation. Following a rejection, both agents return to the pairing stage, and the negotiation resumes only if both choose to attend it.

ternating offers protocol of Rubinstein (1982). The proposer offers a wage w, which the responder either accepts or rejects. If the offer is accepted, an employment relationship is formed, and production begins in the following period. Both agents leave the search market and are replaced by new entrants.

If an offer is rejected, the negotiation can continue into the following period, with the responder becoming the next proposer. Importantly, bargaining does not remove agents from the pairing process. Before the counteroffer is made, both parties return to the pairing stage and may be paired with alternative partners. Figure 1 illustrates the timing of a negotiation following a pairing.

All negotiations take place simultaneously, and an agent can attend at most one of them. Any negotiation that either party does not attend dissolves permanently, and the parties lose contact. An agent can therefore carry at most one ongoing negotiation from the previous period. Since each agent can also be paired with at most one new partner per period, they can have at most two available negotiations: an ongoing one and a newly formed one.

At the end of the pairing stage, after Nature's proposer draws, each agent chooses which

negotiation, if any, to attend. These decisions are simultaneous. A negotiation proceeds to bargaining only if both parties choose to attend it.

2.3 Information and Markov strategies

We restrict attention to Markov strategies, common to all agents of the same type. Under these strategies, agents behave in the same way whenever they face the same bargaining opportunities. At the pairing stage, their decisions depend only on the negotiations currently available to them and their bargaining role in each. At the bargaining stage, wage offers depend only on the identity of the proposer, while acceptance decisions depend on the wage offered. Strategies are independent of calendar time, previously rejected offers, the duration of negotiations, and agents' past relationships.

At the pairing stage, an agent observes their available negotiations and the bargaining role they would occupy in each. For an ongoing negotiation, this role is determined by the previous rejection; for a new negotiation, it is determined by Nature's draw.

Formally, the state of agent i at the pairing stage is

$$s_i = (q_i^o, q_i^n), \qquad q_i^o, q_i^n \in \{P, R, 0\}, \tag{3}$$

where P and R denote proposer and responder, respectively, and 0 indicates that a negotiation is unavailable. The superscripts o and n refer to ongoing and new negotiations.

An attendance strategy specifies which negotiation an agent chooses to attend in each state:

$$a_i(s_i) \in \{o, n, 0\},$$

where the choice set is restricted to available negotiations and 0 denotes attending neither. When an agent is indifferent between attending their only available negotiation and attending none, I adopt the convention that they attend.[1]

Agents observe their own state but not the other negotiations available to their partners. Thus, an agent does not know whether a partner will choose to attend their common negotiation. Attendance decisions take into account both the bargaining role in each negotiation and the probability that the respective partner also attends. Since pairing draws are independent over time and strategies are Markov, an agent's beliefs about their partner's alternatives are given by the pairing probabilities and are never revised along the history of a negotiation.

[1]This allows us to rule out any equilibria in which agents do not attend any negotiations because they expect no one else to attend either, are therefore indifferent and do not attend any meeting.

At the bargaining stage, the state is simply the identity of the proposer. The proposer chooses a wage $w \in \mathbb{R}$ to offer, and the responder decides whether to accept or reject it according to a response rule $r_i(w) \in \{1, 0\}$, where 1 denotes acceptance and 0 rejection.

2.4 Equilibrium characterisation

Optimality of attendance decisions. At the pairing stage, an agent chooses which negotiation, if any, to attend. Let $V_i^a(s_i)$ denote the expected value to agent i of attendance choice a in state s_i. An optimal attendance strategy satisfies

$$a_i^*(s_i) \in \arg\max_{a \in A(s_i)} V_i^a(s_i), \tag{4}$$

where $A(s_i) \subseteq \{o, n, 0\}$ is the set of feasible attendance choices.

Optimal wage offers. The proposer chooses the wage that maximises their expected value, given the responder's response rule. Recall that $V_i(w)$ is the value of an employment relationship at wage w at the start of its first production period. Let $V_i^{\text{rej}}(P)$ and $V_i^{\text{rej}}(R)$ denote the continuation values following a rejection when acting as proposer and as responder, measured at the start of the following period. For proposer i facing responder j,

$$w_i^* \in \arg\max_{w \in \mathbb{R}} \left\{ r_j^*(w) V_i(w) + \big(1 - r_j^*(w)\big) V_i^{\text{rej}}(P) \right\}. \tag{5}$$

Optimal response rule. The responder accepts an offer whenever the value of agreement is at least as large as the value of rejection:

$$r_i^*(w) = \begin{cases} 1, & \text{if } V_i(w) \geq V_i^{\text{rej}}(R), \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

Definition 1 (Markov perfect equilibrium) *A Markov perfect equilibrium consists of attendance strategies $a_i(s_i)$, wage offers w_i, response rules $r_i(w)$, and beliefs about partners' attendance decisions, such that:*

1. *Attendance decisions satisfy* (4).
2. *Wage offers satisfy* (5).
3. *Response rules satisfy* (6).
4. *The rejection values $V_i^{\text{rej}}(P)$ and $V_i^{\text{rej}}(R)$ are the continuation values obtained when all agents follow strategies satisfying* (4),(5) *and* (6) *at every subsequent decision point.*

5. *Beliefs about partners' attendance decisions are consistent with the strategies and the pairing process.*

Proposition 1 (Equilibrium characterisation) *There exists a unique Markov perfect equilibrium, characterised by the following strategies:*

1. ***Bargaining.*** *The firm offers w_f^* when proposing, while the worker offers w_w^*. The worker accepts any wage $w \geq w_f^*$, while the firm accepts any wage $w \leq w_w^*$.*

2. ***Attendance.*** *An agent with only one available negotiation attends it. When both an ongoing and a new negotiation are available, the agent attends the new negotiation, except when they are the proposer in the ongoing negotiation and the responder in the new one.*

   Formally, the equilibrium attendance strategy is

$$a_i^*(s_i) = \begin{cases} o, & s_i \in \{(P,0),(R,0),(P,R)\}, \\ n, & s_i \in \{(0,P),(0,R),(P,P),(R,P),(R,R)\}, \\ 0, & s_i = (0,0). \end{cases} \tag{7}$$

The proof of existence, uniqueness, and the equilibrium characterisation in Proposition 1 is provided in Appendix A.1. Here, I provide the intuition behind this result.

When two parties meet to bargain, they bargain by alternating offers. Delay is costly: future surplus is discounted, and the negotiation may break down if one party leaves for a new partner. A proposer therefore gains nothing from an offer that will be rejected. They make an offer the responder will accept and concede no more than they must. The wage they offer leaves the responder indifferent between accepting now and waiting to propose in the next round, and it is accepted at once. Because waiting is costly, the responder settles for less than they could secure as the proposer, so the proposer takes the larger share. This is the reason why the firm's wage offer is smaller than what the worker would ask: $w_f^* < w_w^*$.

As optimal wage offers make responders indifferent between accepting and rejecting, we need to pin down the equilibrium value of rejection. We therefore consider the equilibrium play after the last round ended with rejection.

In that case, if an agent is paired with two partners, the only two relevant factors are the probability that each partner shows up, and what that agent would be able to obtain if they do. The second consideration favours the partner with whom they would propose, since the proposer's share is the larger one. The first favours the new partner, because an ongoing partner may have left for a new negotiation of their own, while a newly arrived one cannot

have: in equilibrium, all offers are immediately accepted so that no newly paired partner can be in an ongoing negotiation. Hence, an agent who is a responder in an ongoing negotiation, but a proposer in the new one would choose the new negotiation. This strategy induces the proposer in the ongoing negotiation to choose a new negotiation if they are proposer in the new one as well. Therefore, an agent who is a responder in both would choose the new one too. The only case in which the decision is not obvious is if an agent is a proposer in the old, but a responder in the new negotiation. As we will see when we characterise equilibrium values, an agent chooses the ongoing negotiation in this case.[2]

2.5 Equilibrium values and surplus division

Having characterised the equilibrium strategies, we now derive the equilibrium values and the resulting division of surplus. Let the surplus accruing to the worker and the firm from an employment agreement at wage w, relative to continued search, be

$$S_w(w) = V_w(w) - V_w^0, \qquad S_f(w) = V_f(w) - V_f^0. \tag{8}$$

In the generalised Nash bargaining solution, the worker's bargaining weight β determines the division of surplus according to

$$\frac{S_w(w)}{S_f(w)} = \frac{\beta}{1-\beta}. \tag{9}$$

Our aim is to derive this division directly from the non-cooperative bargaining game, rather than imposing β as an exogenous parameter.

Proposition 1 establishes that all equilibrium offers are accepted immediately. The values of unemployment and an open vacancy therefore satisfy

$$\begin{aligned} V_w^0 &= b + \delta\left[(1-\lambda_w)V_w^0 + \frac{\lambda_w}{2}\big(V_w(w_f^*) + V_w(w_w^*)\big)\right], \\ V_f^0 &= -c + \delta\left[(1-\lambda_f)V_f^0 + \frac{\lambda_f}{2}\big(V_f(w_f^*) + V_f(w_w^*)\big)\right]. \end{aligned} \tag{10}$$

These values reflect the probability of being paired with a new partner and the equal probability of making the initial offer.

We next use the responder's indifference conditions to characterise the equilibrium wages and the corresponding surplus shares. Consider a worker receiving the firm's offer w_f^*. Upon

[2]In Appendix A.1, I also show that there is no equilibrium in which a proposer in an ongoing negotiation abandons it in favour of the new negotiation where they are responder.

rejection, the worker becomes the proposer in the ongoing negotiation in the following period. They may also be paired with a new firm, so indifference requires

$$V_w(w_f^*) = b + \delta \Bigg[\frac{\lambda_w}{2} V_w(w_w^*) + \left(1 - \frac{\lambda_w}{2}\right) \left((1-\lambda_f) V_w(w_w^*) + \lambda_f V_w^0\right) \Bigg]. \tag{11}$$

With probability $\frac{\lambda_w}{2}$, the worker is paired with a new firm in which they can propose. In equilibrium, they prefer this negotiation. Otherwise, they prefer the ongoing negotiation. The firm remains available with probability $1-\lambda_f$, the worker gets to make their wage offer which is accepted in equilibrium. With probability λ_f, the firm has found a new worker and left, leaving the worker unemployed.

Similarly, a firm receiving the worker's offer w_w^* is indifferent between accepting and rejecting when

$$V_f(w_w^*) = -c + \delta \Bigg[\frac{\lambda_f}{2} V_f(w_f^*) + \left(1 - \frac{\lambda_f}{2}\right) \left((1-\lambda_w) V_f(w_f^*) + \lambda_w V_f^0\right) \Bigg]. \tag{12}$$

Subtracting the search values in (10) from the indifference conditions (11) and (12), and rearranging, we obtain the individual surplus equations:

$$\begin{aligned} S_w(w_f^*) &= \delta \Bigg[S_w(w_w^*) - \underbrace{\lambda_f \left(1 - \frac{\lambda_w}{2}\right) S_w(w_w^*)}_{\text{Breakdown risk}} \\ &\qquad - \underbrace{\frac{\lambda_w}{2} \left(S_w(w_w^*) + S_w(w_f^*)\right)}_{\text{Outside opportunities}} \Bigg], \\ S_f(w_w^*) &= \delta \Bigg[S_f(w_f^*) - \underbrace{\lambda_w \left(1 - \frac{\lambda_f}{2}\right) S_f(w_f^*)}_{\text{Breakdown risk}} \\ &\qquad - \underbrace{\frac{\lambda_f}{2} \left(S_f(w_f^*) + S_f(w_w^*)\right)}_{\text{Outside opportunities}} \Bigg]. \end{aligned} \tag{13}$$

These equations reveal two channels through which the availability of alternative partners affects bargaining.

The first is the risk that the original negotiation breaks down. Consider the worker's equation. Following rejection, the worker becomes the proposer in the ongoing negotiation, but the original firm may find another worker and leave. This occurs with probability λ_f. However, losing the original firm matters only if the worker has not already switched to a new negotiation in which they can propose, which happens with probability $1 - \frac{\lambda_w}{2}$. A higher probability that the firm finds an alternative therefore reduces the worker's continuation surplus.

The second channel concerns the value of outside opportunities. Following rejection, the worker expects to propose in the next round. A new pairing does not improve on this bargaining position: the worker either obtains an equivalent opportunity to propose elsewhere or prefers to remain with the original firm.

By contrast, a worker without an ongoing negotiation benefits from any new pairing, whether as proposer or responder. The expected surplus from a new pairing is $\frac{1}{2}[S_w(w_w^*) + S_w(w_f^*)]$. A higher pairing probability therefore raises the value of searching. Since the worker's surplus is the value of an employment agreement minus the value of searching, an increase in the latter without a corresponding increase in the former reduces the surplus. The firm's equation has the symmetric interpretation.

These two channels can work in opposite directions. In a tighter labour market, workers are less likely to lose their current bargaining partner, strengthening their position. At the same time, their own outside opportunities become more valuable, reducing the surplus from maintaining a particular negotiation relative to continued search. The equilibrium division of surplus reflects the interaction of these forces.

Define $\beta_f = \frac{S_w(w_f^*)}{S_w(w_f^*)+S_f(w_f^*)}$ and $\beta_w = \frac{S_w(w_w^*)}{S_w(w_w^*)+S_f(w_w^*)}$ as the worker's surplus shares when the firm and worker propose, respectively. Combining the indifference conditions with (10) yields

$$\begin{aligned} \beta_f &= \frac{\delta(1-\lambda_f)(2-\lambda_w)}{2+\delta[1+(1-\lambda_w)(1-\lambda_f)]}, \\ \beta_w &= \frac{2+\delta\lambda_w}{2+\delta[1+(1-\lambda_w)(1-\lambda_f)]}. \end{aligned} \tag{14}$$

The worker obtains a larger share when proposing than when responding, $\beta_w > \beta_f$, reflecting the proposer advantage established earlier. Since workers and firms are equally likely to make the initial offer, the worker's expected bargaining power is

$$\bar{\beta} = \frac{1}{2}(\beta_f + \beta_w) = \frac{1}{2}\left[1 + \frac{\delta(\lambda_w - \lambda_f)}{2+\delta[1+(1-\lambda_w)(1-\lambda_f)]}\right]. \tag{15}$$

First, we can notice that the worker's expected share equals one half when $\lambda_w = \lambda_f$, exceeds one half when $\lambda_w > \lambda_f$, and falls below one half when $\lambda_w < \lambda_f$. Since workers and firms differ only in their pairing probabilities, unequal bargaining power arises from differences in their access to alternative partners. Second, notice that the denominator reflects the frictions in the market. The product $(1 - \lambda_f)(1 - \lambda_w)$ is the probability that neither side is paired with an alternative between rounds, and it governs how far a given gap $\lambda_w - \lambda_f$ moves the split. When the market is more frictional, the fact that one side holds market power matters less. As the market becomes more efficient, so that at least one side is likely to find an alternative between rounds, market tightness matters more for the division of surplus.

Proposition 2 formally establishes that the worker bargaining power increases with tightness.

Proposition 2 (Bargaining power and market tightness) *Worker's equilibrium bargaining power is strictly increasing in labour market tightness:*

$$\frac{\partial \bar{\beta}}{\partial \theta} > 0. \tag{16}$$

The proof is provided in Appendix A.2. A tighter labour market reduces the risk that a worker loses the opportunity to make a counteroffer, but also raises the value of searching relative to maintaining an ongoing negotiation. In the symmetric case, in equilibrium, the former effect dominates.

2.6 Equilibrium wages

Having characterised the equilibrium division of surplus, we now derive the wages agreed upon by workers and firms. Using (1) and (8), the individual surpluses can be expressed as

$$\begin{aligned} S_w(w) &= \frac{w - (1-\delta)V_w^0}{1 - \delta(1-s)}, \\ S_f(w) &= \frac{p - w - (1-\delta)V_f^0}{1 - \delta(1-s)}. \end{aligned} \tag{17}$$

Using the equilibrium bargaining shares β_f and β_w, we obtain the wages offered by the firm and the worker, respectively:

$$\begin{aligned} w_f^* &= \beta_f p + (1-\delta)\left[(1-\beta_f)V_w^0 - \beta_f V_f^0\right], \\ w_w^* &= \beta_w p + (1-\delta)\left[(1-\beta_w)V_w^0 - \beta_w V_f^0\right]. \end{aligned} \tag{18}$$

The expected equilibrium wage is

$$\bar{w} = \bar{\beta}(\theta)p + (1-\delta)\left[(1-\bar{\beta}(\theta))V_w^0 - \bar{\beta}(\theta)V_f^0\right]. \tag{19}$$

The wage equations have two main components. The first term is a share of the total revenue generated by the employment relationship p. The second and third terms reflect how market values of search for both sides affect wages. In particular, we can notice that when worker bargaining power is 0, their wage simply equals their unemployment value. In contrast, when worker bargaining power is 1, they would obtain $p - (1-\delta)V_f^0$, essentially compensating the firm just enough to exit the search market.

In these expressions, we take search values as given. This representation will be particularly useful for the empirical analysis as we will then need to separate between components that are specific to the match and those that are common to the market. Importantly, the wage equations here make explicit the role of endogenous bargaining power in the relationship between match productivity and wages.

Naturally, in the model, the values of continued search are themselves determined in equilibrium. Appendix A.1.2 derives these values, and Appendix A.3 provides the corresponding equilibrium wage expressions after substituting them.

Role of first mover probabilities. In the baseline analysis, we focused on the case in which firms and workers are equally likely to make the first offer, setting μ, firm's first mover probability to $\frac{1}{2}$. This case is instructive because it allows us to isolate market tightness as the only difference between the two sides of the market, without giving any side of the market an advantage. It also allows us to derive tractable expressions for bargaining power in equilibrium.

In practice, workers and firms may differ in their likelihood of making the first offer, as firms may be more likely to make the first offer by posting a wage. In Appendix A.4, I derive the equilibrium surplus shares for any $\mu \in (0,1)$.

In this more general case, the sign of the relationship between bargaining power and tightness is no longer unambiguous. It depends on the relative responsiveness of workers' and firms' pairing probabilities to market tightness, which is determined by the functional form of the pairing technology. These elasticities govern how the two forces shaping bargaining power, the risk of breakdown and the opportunity cost of delaying agreement, respond to changes in labour market tightness.

To investigate this relationship, I consider Cobb–Douglas and CES pairing functions over a range of matching elasticities. The numerical results show that worker bargaining power increases with tightness for a broad range of parameters, including those consistent with

empirical estimates of matching elasticities. I also consider a Leontief pairing technology to illustrate how the relationship can reverse when one side's pairing probability becomes locally unresponsive to tightness. Appendix A.5 presents an analytical example to convey the intuition and numerical exercises in detail. Overall, these results suggest that, although the positive relationship is not guaranteed for every pairing technology, it remains a robust prediction for empirically relevant matching elasticities.

The framework can accommodate different production technologies, vacancy costs, and unemployment values, and can therefore be applied to labour markets with different economic environments. However, the benchmark assumes that firms and workers are homogeneous within each market. While bargaining power can differ across markets as tightness and pairing conditions vary, it is identical across employment relationships within a given market. An important additional source of variation arises when different matches generate different surpluses. I now introduce firm productivity heterogeneity to examine how bargaining power varies within a market, and how these differences interact with labour market tightness.

3 Firm productivity heterogeneity

Let firm productivity p be distributed according to the cumulative distribution function $F(p)$. As in the benchmark, productivity determines the output generated by an employment relationship. I denote the total surplus associated with a firm of productivity p by $S(p)$ and assume that $S'(p) > 0$. All firms generate strictly positive surplus from agreement.

The bargaining protocol and pairing process remain unchanged. In particular, workers and firms continue to be paired with new potential partners while negotiating, the identity of the initial proposer is determined with equal probability, and agents can attend at most one negotiation per period. Pairing probabilities depend on market tightness, but not on firm productivity.

Equilibrium bargaining. The equilibrium is characterised by the same optimality conditions as in the benchmark, laid out in Definition 1. In particular, agents choose which negotiation to attend, which wage to offer, and whether to accept or reject an offer so as to maximise their expected payoffs.

What changes is the information entering these decisions. The state defined in Equation (3) is now augmented by firm productivity. A firm observes its own productivity p, while a worker observes the productivity of the firm in their ongoing negotiation, p, and, when paired with another firm, the productivity p' of the new firm. Attendance decisions therefore depend not only on the bargaining roles and the availability of each partner, but also on the

productivity of the firms involved. Equilibrium wage offers and acceptance decisions likewise depend on the productivity of the firm with which the agent is bargaining.

The introduction of firm heterogeneity changes the comparison between an ongoing and new firm for workers. In the homogeneous model, the bargaining role and the availability of the current partner determine which negotiation a worker prefers. With heterogeneous firms, the productivity of the alternative employer also matters. A worker may prefer to remain in an ongoing negotiation with a highly productive firm even when a new negotiation comes with the role of proposer.

The worker's attendance decisions can therefore be characterised by productivity cutoffs. For each combination of bargaining roles in the ongoing and new negotiations, there is a threshold productivity above which the new firm is sufficiently attractive to induce the worker to leave. These thresholds depend on the productivity of the current firm and are determined jointly with equilibrium wage offers.

Proposition 3 characterises the equilibrium strategies.

Proposition 3 (Equilibrium with heterogeneous firms) *The equilibrium strategies are characterised as follows:*

1. ***Bargaining.*** *At a firm of productivity p, the firm offers $w_f^*(p)$ when proposing, while the worker offers $w_w^*(p)$. The worker accepts any wage $w \geq w_f^*(p)$, while the firm accepts any wage $w \leq w_w^*(p)$. All equilibrium offers are accepted immediately.*

2. ***Attendance.*** *A firm follows the same attendance strategy as in* (7)*.*

   A worker with only one available negotiation attends it. When both an ongoing and a new negotiation are available, the worker's choice depends on their bargaining roles and the productivities of the two firms.

   Let $p^*(p; s_w)$ denote the productivity of a new firm that makes the worker indifferent between the ongoing negotiation with a firm of productivity p and the new negotiation, given the bargaining-role state $s_w = (q_w^o, q_w^n)$ defined in (3)*.*

   The worker's optimal attendance strategy, when both negotiations are available, is

$$a_w^*(s_w; p, p') = \begin{cases} n, & p' \geq p^*(p; s_w), \\ o, & p' < p^*(p; s_w), \end{cases} \tag{20}$$

   where p' is the productivity of the newly encountered firm. When neither negotiation is available, the worker attends neither.

The equilibrium conditions determining the productivity cutoffs are provided in Appendix A.6.

Productivity cutoffs are key to determine the probability with which worker leaves after a rejection. Naturally, with these equilibrium strategies, this probability will be different for two firms with different productivities. Corollary 1 establishes that productivity cutoff is increasing in firm's own productivity.

Corollary 1 (Productivity and switching cutoffs) *For every state $s_w \in \{P, R\}^2$, the worker's switching cutoff is strictly increasing in the incumbent firm's productivity: for any $p' > p$,*

$$p^*(p'; s_w) > p^*(p; s_w) \tag{21}$$

The proof is provided in Appendix A.6.

Workers bargaining with more productive firms therefore become more selective about alternative employers. Although they encounter new firms at the same rate, a smaller fraction of those firms is sufficiently attractive to induce them to abandon their current negotiation.

Implications for bargaining power. The probability that a firm abandons an ongoing negotiation depends on its bargaining role and market tightness, but not on its productivity. By contrast, the probability that a worker abandons an ongoing negotiation depends on the productivity of the incumbent firm.

For a worker whose role in the ongoing negotiation is $m \in \{P, R\}$, this probability is

$$\Pr(\text{worker leaves} \mid p, m, \theta) = \frac{\lambda_w(\theta)}{2} \sum_{r \in \{P,R\}} \bar{F}\left(p^*(p; (m, r))\right), \tag{22}$$

where $\bar{F} = 1 - F$.

By Corollary 1, this probability is decreasing in the incumbent firm's productivity. Workers negotiating with more productive firms are therefore less likely to abandon their ongoing negotiations. This creates a force towards lower worker bargaining power at more productive firms.

The equilibrium expressions for bargaining power, derived in Appendix A.6, are more complex than in the benchmark. Bargaining shares are determined jointly with the endogenous productivity cutoffs and do not generally admit simple closed-form solutions. The effects of firm productivity and market tightness on bargaining power can therefore not be signed analytically in general.

3.1 Numerical analysis

I therefore turn to a numerical analysis to examine how worker bargaining power varies across the firm productivity distribution and how its response to labour market tightness differs between less and more productive firms. I solve the model numerically under the following parameterisation.

Table 1: Parameter choices for the numerical illustration

Parameter	Choice
Pairing function	$\lambda_w(\theta) = \rho\theta^{\frac{1}{2}}, \qquad \lambda_f(\theta) = \rho\theta^{-\frac{1}{2}}$
Pairing efficiency	$\rho = 0.25$
Firm productivity distribution	$p = \exp(0.8z), \qquad z \sim \text{Beta}(2, 1)$
Discount factor	$\delta = 0.99$
Separation probability	$s = 0.02$
Unemployment payoff	$b = 0.25$
Vacancy cost	$c = 0.05$
Firm first mover probability	$\mu = \frac{1}{2}$
Tightness values compared	$\theta \in \{0.5, 1.5\}$

Figure 2 plots expected worker bargaining power across the productivity distribution for the two values of market tightness.

There are three main takeaways from the figure. First, worker bargaining power tends to decline with firm productivity. This is consistent with the mechanism developed above: workers bargaining with more productive firms are less likely to abandon the negotiation, which weakens the role of alternative employers. Second, the effect of tighter labour markets is stronger for less productive firms. Raising market tightness from $\theta = 0.5$ to $\theta = 1.5$ increases worker bargaining power substantially at the lower end of the productivity distribution, but this effect becomes much smaller at higher productivity levels and eventually reverses.

This reversal reflects the two competing channels identified in the benchmark. A tighter labour market reduces the probability that the incumbent firm leaves, increasing the worker's value of continuing negotiations. However, it also raises the value of unemployment by making alternative employers easier to find. At the top of the productivity distribution, few alternative firms are sufficiently attractive to induce the worker to abandon the ongoing negotiation. Consequently, more frequent pairings contribute relatively little to the value of continuing that negotiation, even as they increase the value of unemployment. In the numerical example, this second channel becomes sufficiently important to offset the reduction

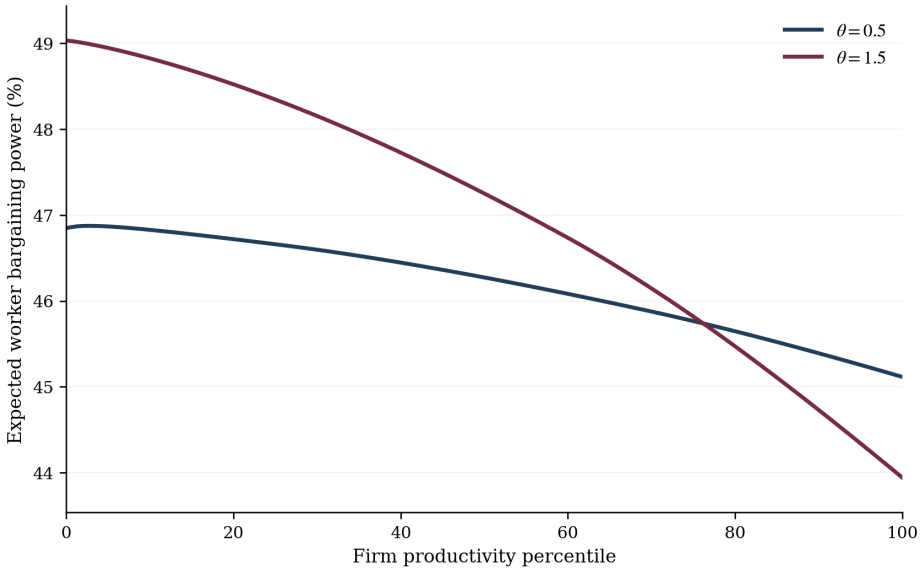


Figure 2: Worker bargaining power across the firm productivity distribution. The horizontal axis reports firm productivity percentiles, while the vertical axis reports average worker bargaining power $\bar{\beta}(p, \theta)$ expressed as a percentage of total match surplus. The blue line corresponds to a labour market with tightness $\theta = 0.5$, and the burgundy line to a tighter market with $\theta = 1.5$. Both series are obtained from the numerical equilibrium using the parameterisation reported in Table 1.

in breakdown risk, leading worker bargaining power to decline with tightness at the most productive firms.

Finally, the relationship between pass-through and firm productivity is much flatter when the markets are less tight. Intuitively, this is because in such markets, it is unlikely for workers to find any other alternative firm, regardless of the firm's productivity.

Empirical counterpart of bargaining power. To take the model to the data, I measure worker bargaining power through the pass-through of firm productivity into wages. Recall from (19) that the expected equilibrium wage in the benchmark is

$$\bar{w} = \bar{\beta}(\theta)p + (1-\delta)\left[(1-\bar{\beta}(\theta))V_w^0 - \bar{\beta}(\theta)V_f^0\right]. \tag{23}$$

This expression provides a direct empirical counterpart to worker bargaining power: the coefficient on firm productivity captures the share of productivity passed through to wages, while the remaining terms reflect the values of continued search. The wage equation can therefore be taken to the data using a linear regression of wages on firm productivity, provided that the empirical specification adequately controls for differences in search values. I discuss the choice of controls and the corresponding identification strategy in the empirical section.

Productivity pass-through is also a commonly used measure of workers' ability to capture firm surplus in the empirical rent-sharing literature (Card et al., 2018). In particular, Mengano (2022) uses wage pass-through to measure workers' bargaining power within a Nash bargaining framework. I follow this approach, interpreting the estimated pass-through as an empirical measure of worker bargaining power.

The benchmark predicts that bargaining power, and consequently productivity pass-through, increases with labour market tightness. The heterogeneous firm extension motivates two further questions: how productivity pass-through varies across the firm productivity distribution, and whether its sensitivity to labour market tightness differs between less and more productive firms.

Having established the model's empirical predictions, I now turn to the data used to test them. I first describe the data sources and construction of the key variables, before presenting the empirical strategy.

4 Data

The empirical analysis combines three sources of French administrative data. Worker wages and employment characteristics come from the employee records in the *Base Tous Salariés* (BTS–Postes). These records are linked to firm balance-sheet information from FARE, which I use to estimate industry-specific production functions. I combine these data with measures of local labour market tightness constructed by the French Ministry of Labour (DARES).

Together, these sources provide the three main objects used in the empirical analysis: workers' hourly wages, the marginal product of labour at their employer, and the tightness of their local labour market. The firm data also allow me to locate each employer in the productivity distribution of comparable firms, which I use to study the heterogeneous firm implications of the model.

4.1 Matched Employer-Employee data

BTS Postes contains administrative records covering employees in France. I use annual files for 2011–2023, the years for which the labour market tightness measure described in Section 4.2 is available.

The unit of observation is a job, defined as the employment of a given worker in a given establishment during a calendar year. For each job, the data report gross and net annual pay, total hours worked, contract type, a four-digit occupation code, and the worker's age and gender. I also observe the location and industry of the employing establishment and the

identifier of the firm to which it belongs.

Using the firm identifier, I link BTS Postes to FARE, a firm-level accounting database covering private-sector firms, excluding financial enterprises and firms in agriculture. FARE provides information on firms' industries, value added, turnover, fixed assets, and intermediate inputs.

Panel creation. FARE provides longitudinal firm records linked by a stable firm identifier. By contrast, BTS Postes consists of repeated cross-sections in which worker identifiers change across annual files. Each file, however, covers two years: it reports each worker's jobs in the current year and, where applicable, the preceding year. A job held in a given year can therefore appear in two consecutive files. Following Babet et al. (2025), I exploit this overlap to link workers across files, matching on job characteristics such as the employer, gender, hours worked, and employment-spell dates. Chaining these links produces a longitudinal worker panel spanning years and employers.

4.1.1 Labour productivity

The first empirical exercise measures worker bargaining power through the pass-through of the firm's marginal product of labour into wages. I construct the marginal product of labour from production functions estimated using the FARE accounts.

For firm j in year t, I assume a Cobb–Douglas value-added production function,

$$y_{jt} = \beta_\ell \ell_{jt} + \beta_k k_{jt} + \omega_{jt} + \varepsilon_{jt}, \tag{24}$$

where y_{jt} denotes log real value added, ℓ_{jt} log labour input measured in full-time-equivalent employees, k_{jt} log real capital, ω_{jt} firm productivity, and ε_{jt} measurement error.

Nominal value added, capital, and intermediate inputs are converted into real terms using industry-specific deflators from EU KLEMS.[3] The value-added and intermediate-input deflators are extended to 2022 and 2023 using industry-level price developments from Eurostat. All price indices are expressed relative to a common base year, 2020.

Firms choose their inputs in response to productivity shocks observed by the firm but not by the econometrician, making OLS estimates of the production function inconsistent. I therefore estimate the production function following Levinsohn and Petrin (2003), using intermediate inputs as a proxy for unobserved productivity. The production-function parameters are estimated separately for each two-digit NACE industry, allowing output elasticities

[3]The EU KLEMS database and its documentation are available at `https://euklems-intanprod-llee.luiss.it/`.

to differ across industries.

Before estimation, I exclude observations with missing or nonpositive production variables and winsorise value added, capital, labour, and intermediate inputs at the first and ninety-ninth percentiles within each industry-year cell. I retain industry-year cells with at least 100 observations.

Given the estimated labour elasticity for industry s, I construct the hourly marginal product of labour for firm j as

$$\mathrm{MPL}_{jt} = \widehat{\beta}_{\ell,s} \frac{Y_{jt}}{H_{jt}}, \tag{25}$$

where Y_{jt} denotes real value added and H_{jt} total hours worked at the firm. Although labour input in the production function is measured in full-time-equivalent employees, I express the marginal product per hour to make it comparable to the hourly wages observed in the worker data. I use this measure to estimate how changes in the productive contribution of labour are transmitted into wages. Productivity pass-through is the empirical measure of worker bargaining power used throughout the analysis.

4.1.2 Firm productivity

The heterogeneous firm analysis requires a separate measure of the productivity of the firm with which the worker is bargaining. I do not use the level of MPL for this purpose. MPL is the productive margin whose pass-through into wages is used to measure bargaining power, whereas the heterogeneous model asks how this pass-through differs across firms with different underlying productivities.

I recover firm productivity from the same industry-specific Levinsohn–Petrin production-function estimates. Because firm-specific output prices are not observed, I construct a revenue-based measure of total factor productivity (TFPR) using real value added. This measure captures differences in firm performance after accounting for the contributions of labour and capital, but may reflect differences in output prices and demand as well as productive efficiency. Specifically, I calculate

$$\widehat{\omega}_{jt} = y_{jt} - \widehat{\beta}_{\ell,s} \ell_{jt} - \widehat{\beta}_{k,s} k_{jt}. \tag{26}$$

Rather than comparing TFPR levels across industries, I use firms' relative positions in their industry-specific productivity distributions. To reduce the influence of temporary fluctuations in measured productivity, I construct rankings over three periods: 2011–2014, 2015–2018, and 2019–2021. The final period ends in 2021 because the capital deflators used to construct TFPR are not available for 2022 and 2023.

Within each two-digit NACE industry and year, I calculate firms' percentile ranks in the TFPR distribution. I then average these annual ranks within each period, requiring at least two valid productivity observations per firm.

Based on these average ranks, I assign firms to productivity deciles using two alternative reference distributions. National deciles are constructed within each industry and period, whereas regional deciles are constructed within each industry, region, and period. For the regional classification, a firm's region is determined by its location in the last year in which it is observed within the period.

Productivity classifications remain fixed within each period, with the 2019–2021 classification also applied to 2022 and 2023. Firms may change productivity groups between periods.

MPL therefore provides the variation used to estimate wage pass-through, while TFPR rank captures the productivity of the employer relative to comparable firms. I then examine how productivity pass-through and its sensitivity to labour market tightness vary across this distribution.

4.1.3 Sample restriction

I restrict attention to the main job held by each worker, as flagged in the data, and construct hourly wages using gross annual pay and total hours worked. I include open-ended and fixed-term employment contracts, excluding apprentices, trainees, temporary agency workers, and subsidised jobs, and drop records with zero pay or hours.

I further restrict the sample to firms with at least five full-time-equivalent employees. Since production and value added are observed at the firm level rather than separately for each establishment, I exclude firms operating establishments across multiple administrative areas. This restriction ensures that firm-level productivity can be assigned to a single geographic area, even when a firm operates multiple establishments.

Finally, I exclude observations in the bottom and top 1% of the hourly MPL distribution.

4.2 Labour market tightness

I measure labour market tightness using the indicator constructed annually by DARES, the statistical office of the French Ministry of Labour. The measure varies by occupation, geographic area, and year. Occupations are defined according to the FAP classification (*Familles Professionnelles*), which groups occupations into broad occupational families. Geographic areas correspond to French *départements*, administrative divisions that are smaller than regions and broadly comparable to counties.

I define a local labour market as an occupation–administrative-area pair and denote its tightness in year t by θ_{olt}, where o indexes occupations and l indexes administrative areas. I assign workers to these labour markets using the official correspondence between the occupational classification recorded in BTS Postes and FAP, provided by DARES.[4]

The indicator combines information from vacancy and unemployment registries, flows out of registered unemployment, and employer surveys on recruitment difficulties. Higher values correspond to labour markets in which workers find jobs more easily and firms face greater difficulty filling vacancies. It therefore provides an empirical measure of the relative availability of outside options on the two sides of the market, which is the object captured by tightness in the model. The indicator is standardised over the period 2011–2022.

5 Empirical strategy

The empirical analysis examines two main implications of the theory. First, I test whether productivity pass-through into wages increases with labour market tightness. In the model, the sensitivity of wages to match productivity reflects the worker's bargaining power. The central prediction is therefore that workers capture a larger share of productivity in tighter labour markets.

I then examine the implications of firm heterogeneity, asking whether productivity pass-through and its sensitivity to tightness vary across the firm productivity distribution.

5.1 Productivity pass-through and labour market tightness

I estimate the relationship between wages, the firm's marginal product of labour, and local labour market tightness. For worker i employed at firm j, in occupation o, location l, and year t, the estimating equation is

$$w_{ijolt} = \alpha_1 \,\mathrm{MPL}_{jt} + \alpha_2 \,\theta_{olt} + \alpha_3 \,(\mathrm{MPL}_{jt} \times \theta_{olt}) + X_{it}'\gamma + \phi_j + \phi_o + \phi_l + \phi_t + \varepsilon_{ijolt}. \tag{27}$$

The coefficient α_1 measures productivity pass-through at the reference level of tightness, while α_2 captures the relationship between tightness and wages at the reference level of MPL. The coefficient of primary interest is α_3, which measures how productivity pass-through changes with labour market tightness. In particular,

$$\frac{\partial w_{ijolt}}{\partial \mathrm{MPL}_{jt}} = \alpha_1 + \alpha_3 \theta_{olt}. \tag{28}$$

[4]The correspondence between PCS-2003 and FAP-2009 is available from DARES at `https://dares.travail-emploi.gouv.fr/donnees/la-nomenclature-des-familles-professionnelles-2009`.

The central prediction of the baseline model is therefore $\alpha_3 > 0$.

The vector X_{it} contains worker age, age squared, and gender. These characteristics may influence wages both through worker productivity and through the value of unemployment, which enters the theoretical wage equation. I include occupation and location fixed effects to account for persistent differences in wages and labour market conditions across occupations and locations. Year fixed effects absorb aggregate shocks common to all labour markets. Finally, firms may sort into tighter or slacker labour markets in ways that are correlated with their productivity and wage-setting practices. I therefore include firm fixed effects in the preferred specification, removing persistent differences across employers. Identification then relies on changes in labour market tightness and productivity within firms over time, rather than on differences between firms operating in systematically different labour markets.

5.2 Identification

Estimating Equation (27) by OLS may not recover the causal relationship between productivity pass-through and labour market tightness. I discuss the threats to identification and the instrumental variable strategy for each of the coefficients separately.

Marginal product of labour. The marginal product of labour may be endogenous for several reasons. First, firms' employment decisions affect MPL through diminishing marginal returns to labour. Since wages also influence firms' employment decisions, changes in wages may affect measured MPL through adjustments in labour input. Second, unobserved shocks to firm profitability or labour demand may simultaneously affect MPL and wages, generating omitted-variable bias. Finally, MPL is constructed from firm accounting data and estimated production functions, making it subject to measurement error.

I instrument current MPL using its lagged value, $\text{MPL}_{j,t-1}$, following an approach used in the wage–productivity and rent-sharing literature (Card et al., 2018; Bell et al., 2022). The instrument exploits the persistence of firm productivity and can partially address contemporaneous simultaneity and transitory measurement error. Identification requires that lagged productivity affects current wages only through current productivity, conditional on the included controls and fixed effects.

Labour market tightness. A key issue in identification of tightness effect is that labour market tightness itself is an equilibrium outcome. In general equilibrium, wages, hiring, and vacancy creation are jointly determined. In particular, higher wages tend to discourage vacancy creation, thereby reducing labour market tightness. This creates a source of downward bias in the estimated relationship between wages and tightness.

I therefore instrument local labour market tightness using a lagged leave-one-out measure constructed from tightness for the same occupation in other locations, following a strategy used in the literature on local labour market power (Azar et al., 2022; Marinescu et al., 2021).

Writing a local labour market as $m = (o, l)$, the instrument is

$$\theta^{\text{LOO}}_{m,t-1} = \frac{\sum_{l' \neq l} e_{ol',t-1} \theta_{ol',t-1}}{\sum_{l' \neq l} e_{ol',t-1}}, \tag{29}$$

where $e_{ol',t-1}$ denotes employment in occupation o and location l' in the previous year.

The instrument exploits the common shocks to an occupation that could shift its demand and that are unrelated to the wages in a particular location. Excluding the worker's own location removes its direct contribution to the instrument, while using the previous year's tightness rules out reverse causality from current wage setting to the instrument.

One possible concern is that wages may adjust gradually to changes in labour market conditions. Consequently, past tightness may continue to affect current wages independently of contemporaneous tightness. To alleviate this concern, I repeat the analysis using workers whose employment relationship with their current firm began after 2011, allowing me to identify labour market tightness at the time of hiring. In this specification, I use tightness in the hiring year, instrumented by the leave-one-out measure from the preceding year. Since the instrument predates the worker's employment relationship with the firm, it cannot be affected by wage agreements within that relationship. This reduces concerns that the estimates reflect the persistence of previously negotiated wages rather than labour market conditions at the time of bargaining.

Interaction between productivity and tightness. The interaction between productivity and labour market tightness may also be endogenous, although endogeneity of the individual components does not necessarily imply bias in the interaction coefficient. The specification allows productivity and tightness to affect wages separately; the interaction captures whether the sensitivity of wages to productivity changes with labour market tightness.

A particular concern is that firms' productivity and employment decisions may jointly affect wages and the labour market conditions they face. This is especially relevant for employers accounting for a substantial share of recruitment in their local labour market. A productivity shock at such a firm may raise wages and induce additional hiring, thereby also changing local tightness.

I instrument the interaction using the product of the instruments for MPL and tightness,

$\text{MPL}_{j,t-1} \times \theta^{\text{LOO}}_{m,t-1}$.

The leave-one-out construction excludes the firm's own location, limiting the influence of its hiring decisions on the instrument, while lagging rules out reverse causality from current wage and employment decisions to the instruments.

The resulting two-stage least squares specification uses three excluded instruments: lagged MPL, lagged leave-one-out tightness, and their interaction. These jointly instrument current MPL, current tightness, and their interaction. Identification additionally requires that the interaction of the instruments is uncorrelated with unobserved determinants of current wages, conditional on the controls and fixed effects.

Standard errors are two-way clustered by firm and local labour market, defined by occupation and location. I assess the relevance of the instruments using conditional first-stage diagnostics.

6 Results

6.1 Productivity pass-through and labour market tightness

Table 2 presents the baseline two-stage least squares estimates for the full economy. Column (1) includes occupation, location, and year fixed effects, while column (2) additionally includes firm fixed effects.

The central prediction of the model is that productivity pass-through increases with labour market tightness.

The main coefficient of interest is the interaction between productivity and labour market tightness. It is positive and statistically significant in both specifications, consistent with the central prediction of the model that workers capture a larger share of productivity increases in tighter labour markets. In column (1), a one-standard-deviation increase in tightness raises productivity pass-through by approximately 17% relative to its level in the average labour market. The corresponding increase is approximately 13% when firm fixed effects are included in column (2).

The estimated level of productivity pass-through also differs across specifications. At average tightness, an additional euro of hourly MPL is associated with approximately 7.5 cents in additional hourly wages without firm fixed effects, compared with 3.3 cents when firm fixed effects are included. This reduction is consistent with persistent differences across firms accounting for part of the observed relationship between productivity and wages.

The distinction between these estimates is important for their interpretation. In the model, bargaining power determines how the surplus generated by a match is divided between

the worker and the firm. The empirical counterpart of this object is the pass-through of productivity into wages, including differences in the surplus available across firms. However, the objective of the empirical analysis is not only to measure average pass-through, but to identify how labour market tightness affects it. Firm fixed effects absorb persistent differences in wage levels across employers, while occupation, location, and year fixed effects account for other systematic sources of wage variation. The resulting estimates therefore exploit variation conditional on these characteristics, helping to isolate the relationship between tightness and the share of productivity received by workers.

Table 2: Productivity pass-through and labour market tightness

	Hourly wage	
	(1) No firm FE	(2) Firm FE
Marginal product of labour	0.0749***	0.0329***
	(0.0042)	(0.0052)
Labour market tightness	0.0130	0.0112
	(0.0702)	(0.0435)
MPL × Tightness	0.0129***	0.0042***
	(0.0027)	(0.0012)
Worker controls	Yes	Yes
Occupation FE	Yes	Yes
Location FE	Yes	Yes
Year FE	Yes	Yes
Firm FE	No	Yes
Observations	39,282,927	39,282,927
First-stage Wald F-statistics		
MPL	295.89	140.82
Tightness	2,150.30	2,480.70
MPL × Tightness	754.14	763.01

Notes. The table reports two-stage least squares estimates for the full sample. The dependent variable is the worker's gross hourly wage. MPL is the employer's estimated hourly marginal product of labour. Labour market tightness is the DARES indicator defined at the occupation × location level.
MPL and tightness are centred before constructing their interaction. Current MPL, tightness, and their interaction are instrumented using lagged firm MPL, lagged employment-weighted leave-one-out tightness for the same occupation in other locations, and the product of these instruments, respectively. Worker controls include age, age squared, and gender. Standard errors, reported in parentheses, are two-way clustered by firm and local labour market (occupation × location).
The bottom panel reports first-stage Wald F-statistics for each endogenous regressor, using the same clustering structure.
* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

The coefficient on MPL in the specification with firm fixed effects is 0.0329. At the reference level of tightness, an additional euro of hourly MPL is therefore associated with approximately 3.3 cents in additional hourly wages. The positive interaction implies that this pass-through increases as the labour market becomes tighter.

The coefficient on tightness itself is also positive in the firm fixed effects specification. However, in the presence of the interaction, this coefficient measures the wage response to tightness at the reference level of MPL, rather than an unconditional wage effect. The interaction is therefore the more direct empirical counterpart of the model's central prediction.

Overall, the estimates indicate that workers receive a larger share of marginal increases in their employer's productivity when labour markets are tighter.

6.2 Labour Market Tightness at Hiring

I next investigate whether labour market conditions at the time of hiring are particularly important for wage determination. This exercise is motivated by two considerations. First, measuring labour market tightness at hiring better captures the bargaining conditions prevailing when the employment relationship begins. Second, the theoretical model describes wage bargaining when a new worker–firm match is formed. Labour market tightness at the time of hiring therefore provides a more direct empirical counterpart to the bargaining environment in the model.

To conduct this analysis, I restrict the sample to workers with a recorded hiring event after 2011, when labour market tightness becomes observable. This restriction produces a different subsample, consisting of workers who have experienced at least one recorded job move during the observation period. Table 3 compares the characteristics of the two samples. Workers in the hiring-date sample earn somewhat higher wages, are slightly younger, and face considerably tighter labour markets on average. They also work at firms with higher measured labour productivity. Table 4 then presents the estimation results, comparing specifications using contemporaneous tightness with those using tightness at hiring, both with and without firm fixed effects.

Table 3: Descriptive Statistics by Sample

	Full sample	Hiring-date sample
Hourly wage (€)	18.67	19.77
Marginal product of labour	0.088	0.383
Labour market tightness	0.103	0.305
Age	40.22	39.34
Female (%)	39.03	37.46
Observations	39,287,991	6,516,075

Notes: The table reports sample means. The hiring-date sample consists of workers with at least one recorded hiring event after 2011, allowing labour market tightness to be measured at hiring. Marginal product of labour refers to the demeaned measure used in the empirical analysis.

Table 4: Productivity Pass-Through and Labour Market Tightness at Hiring

	(1) Contemporaneous	(2) At hiring	(3) Contemporaneous	(4) At hiring
Marginal product of labour	0.0787***	0.0783***	0.0368***	0.0367***
	(0.0052)	(0.0060)	(0.0075)	(0.0077)
Market tightness	0.0493		-0.0163	
	(0.0793)		(0.0562)	
MPL × market tightness	0.0069*		-0.0003	
	(0.0035)		(0.0015)	
Tightness at hiring		0.2792***		0.3082***
		(0.0630)		(0.0308)
MPL × tightness at hiring		0.0148***		0.0030*
		(0.0037)		(0.0015)
Worker controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Occupation FE	Yes	Yes	Yes	Yes
Location FE	Yes	Yes	Yes	Yes
Firm FE	No	No	Yes	Yes
Observations	6,516,075	6,516,075	6,489,935	6,489,935
R^2	0.52780	0.52812	0.62495	0.62511

Notes: The dependent variable is gross hourly wages. MPL denotes marginal product of labour. The sample is restricted to observations for workers hired after 2011, for whom labour market tightness at hiring can be identified. Columns (1) and (3) measure tightness contemporaneously with wages, while columns (2) and (4) measure tightness at the time of hiring. All specifications include worker controls and year, occupation, and department fixed effects. Columns (3) and (4) additionally include firm fixed effects. Standard errors in parentheses are clustered at the firm and labour-market levels. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

The results suggest that labour market conditions at hiring play a particularly important role in wage determination. Without firm fixed effects, the coefficient on tightness at hiring is 0.279, compared with 0.049 for contemporaneous tightness. With firm fixed effects, the corresponding coefficients are 0.308 and -0.016, respectively. While contemporaneous tightness is not statistically significant in either specification, tightness at hiring is positive and statistically significant in both. This pattern suggests that wages retain an important imprint of labour market conditions prevailing when the employment relationship begins, consistent with persistent effects of initial bargaining conditions on subsequent wages. These results suggest that labour market conditions at hiring continue to matter for wages throughout the employment relationship, rather than wages adjusting fully to contemporaneous labour market conditions.

The interaction between productivity and labour market tightness provides further evidence of the importance of conditions at hiring. Without firm fixed effects, both interactions are positive and statistically significant, but the coefficient on tightness at hiring is more than twice as large as that on contemporaneous tightness. With firm fixed effects, the interaction with contemporaneous tightness becomes statistically insignificant, whereas the interaction with tightness at hiring remains positive and statistically significant. These results suggest that labour market conditions at the time of hiring play a more important role in determining how productivity gains are passed through to wages, even within firms.

The absence of a significant interaction with contemporaneous tightness in this subsample contrasts with the positive relationship estimated in the full sample. One possible explanation concerns differences in the distribution of labour market tightness. Workers in this subsample may face, on average, tighter labour markets than workers in the full sample since they have actually been able to move across jobs. The numerical results of the model suggest that the effect of tightness on bargaining power can be stronger in relatively slack labour markets and diminish as tightness increases. The weaker contemporaneous interaction in this subsample could therefore be consistent with the relationship predicted by the model.

6.3 Productivity pass-through across firms

The heterogeneous firm extension of the model suggests that the share of productivity gains passed through to wages can depend on firm productivity. To examine this heterogeneity, I re-estimate Equation (27) separately across regional TFPR deciles within broad sectors, using the same instrumental variable strategy as in the baseline analysis. I conduct this exercise within sectors because firms in the same productivity decile across different sectors may differ substantially in their revenues, wages, and attractiveness as potential employers.

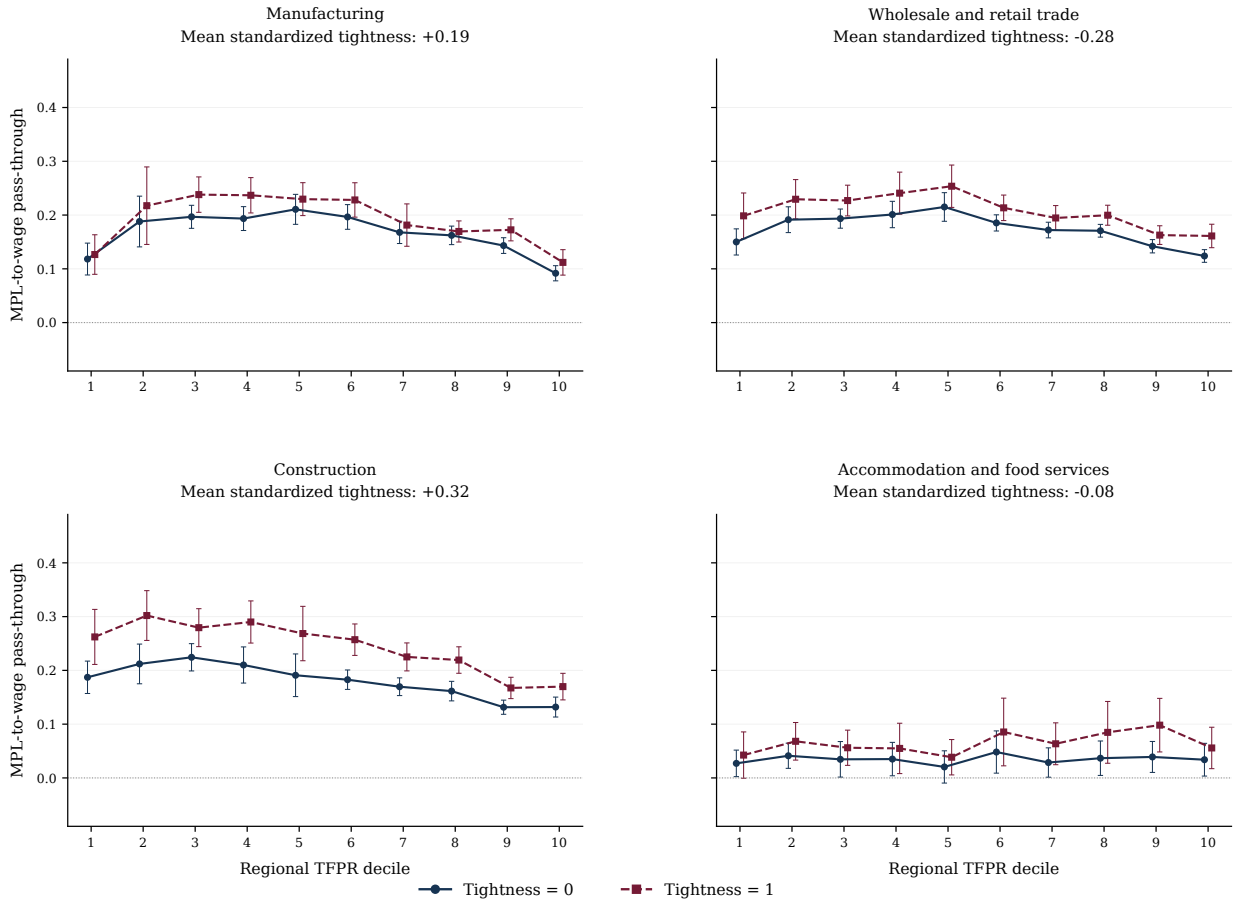


Figure 3: Productivity pass-through across firms: selected sectors

Notes: Each panel reports the estimated pass-through of marginal product of labour (MPL) to hourly wages across regional TFPR deciles within the indicated sector. Pass-through is evaluated at standardized labour market tightness equal to zero (solid navy line) and one (dashed burgundy line). Vertical bars denote 95% confidence intervals. Panel headings report the observation-weighted sector mean of standardized tightness. Regressions are estimated separately by decile.

While the regressions are estimated within broad sectors, TFPR rankings are constructed within two-digit NACE industries and regions, as explained in Section 4.

Unlike the preferred pooled specification, these regressions exclude firm fixed effects to retain variation across firms within productivity deciles. I evaluate the implied productivity pass-through at standardized labour market tightness equal to zero and one.

Figure 3 presents results for manufacturing, wholesale and retail trade, construction, and accommodation and food services. These are the four largest sectors in the estimation sample and together account for approximately 75% of observations. The remaining ten sectors, which individually have smaller samples and often less precisely estimated profiles, are reported in Appendix A.7.

The navy line shows productivity pass-through at standardized labour market tightness

equal to zero. Its shape and level along the productivity distribution differs remarkably across the sectors. In manufacturing and construction, pass-through first increases and then declines towards the top of the productivity distribution. A similar, though less pronounced, pattern emerges in wholesale and retail trade, while the profile is considerably flatter in accommodation and food services. A related non-monotonic relationship between firm productivity and labour markdowns has been documented for US manufacturing by Yeh et al. (2022). Interestingly, these differences are also qualitatively consistent with the numerical results in Figure 2, where bargaining power declines more steeply with firm productivity in a tight labour market, while the relationship is flatter in a slack market. Manufacturing and construction are characterised by higher average tightness, whereas workers in trade and accommodation and food services face relatively slacker markets.

The dashed burgundy line shows productivity pass-through at standardized labour market tightness equal to one. Especially in manufacturing and construction, we see that the effect of one standard deviation increase in tightness is smaller at the top deciles of the productivity distribution. In trade and accommodation and food services on the other hand, the effect of an increase is very similar across the entire distribution, consistent with the flatter relationship between productivity and pass-through.

7 Conclusion

This paper asks what determines worker bargaining power and, in particular, how it depends on labour market conditions. I develop a non-cooperative model of wage bargaining in which workers and firms alternate offers while continuing to search for alternative partners. Following a rejected offer, either party may encounter a new potential partner and decide whether to abandon the ongoing negotiation. These choices make the risk of bargaining breakdown endogenous and dependent on the availability of outside opportunities. Bargaining power therefore emerges as an equilibrium outcome rather than being imposed through an exogenous bargaining weight.

In the benchmark model, the worker's share of match surplus is strictly increasing in labour market tightness. Tightness works through two channels that pull in opposite directions. On one side, firms in tighter markets are less likely to find workers and abandon ongoing negotiations, so a worker who rejects an offer is more likely to retain the opportunity to make a counteroffer. This raises the value of rejecting and, through the firm's offer, the value of agreement. On the other side, tighter markets also raise the value of continued search, since an unemployed worker is paired with firms more often. The surplus from an agreement is what the worker gains over continued search, so a higher value of search leaves

less to be gained from any given agreement, and this works against the first channel. In equilibrium, the first channel dominates: the reduction in breakdown risk raises the value of agreement by more than the value of search rises. The model consequently predicts stronger productivity pass-through into wages in tighter labour markets.

The empirical analysis supports this prediction. Using French matched employer–employee data, I measure worker bargaining power as the pass-through of the firm's marginal product of labour into wages. To address the endogeneity of local labour market conditions, I instrument tightness using the leave-one-out tightness of the same occupation in other locations. In the preferred specification with firm fixed effects, a one-euro increase in the firm's hourly marginal product of labour raises the worker's hourly wage by approximately 3.3 cents at average labour market tightness. A one-standard-deviation increase in tightness raises this pass-through to 3.7 cents per euro, an increase of approximately 13%.

I extend the model to allow firms to differ in productivity and, consequently, in the surplus generated by an employment relationship. Workers negotiating with more productive firms require more attractive alternatives before abandoning their current negotiations. This creates a force towards lower worker bargaining power at more productive firms and can weaken the effect of labour market tightness. The empirical analysis examines these relationships across productivity deciles within sectors. The results reveal substantial heterogeneity in productivity pass-through, both across firms and across sectors. In manufacturing and construction, pass-through initially rises across the lower productivity deciles before generally declining towards the top of the distribution, while the profiles are flatter in trade and accommodation and food services. The effect of labour market tightness on pass-through also varies across productivity deciles, suggesting that the relationship between outside opportunities and wage determination depends on firm productivity and sectoral characteristics.

The model isolates this bargaining mechanism in a simple environment, but the framework can be extended to incorporate multi-worker firms, more general production technologies, on-the-job search, and additional sources of heterogeneity.

More broadly, the results suggest that differences in worker bargaining power across firms and labour markets need not be imposed through an exogenous parameter. They can emerge from the opportunities available to workers and firms, and from the value of those opportunities relative to the negotiations they are already in. One implication is that rent-sharing elasticities estimated in different markets or periods are not directly comparable without accounting for labour market conditions. Labour market conditions shape wages not only by changing what workers and firms can obtain elsewhere, but also by changing how they divide what they create together.

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A Appendix

A.1 Proof of Proposition 1

A.1.1 Equilibrium characterisation

I write $\lambda_i = \lambda_i(\theta)$ for $i \in \{w, f\}$. We work within the class of Markov strategies defined in Section 2.3, with the convention that an agent attends their only available negotiation when indifferent. Let V_i^0 denote the value of remaining in the search market without an ongoing negotiation. The value of agreement at wage w is $V_i(w)$, as in the main text. Both $V_i(w)$ and V_i^0 are evaluated at the start of a period, consistent with employment beginning in the period after agreement. For any pair of types $i, j \in \{w, f\}$, $j \neq i$, agent i is the proposer when the agreed wage is w_i^* and the responder when it is w_j^*.

Throughout, I take as given that gains from agreement are strictly positive in both roles, $V_i(w_i^*) > V_i^0$ and $V_i(w_j^*) > V_i^0$ for $i \in \{w, f\}$. Section A.1.2 verifies that this holds in the equilibrium constructed there, as a consequence of $p - b + c > 0$.

Lemma 1 (Bargaining strategies) *Conditional on both agents attending, the proposer offers the wage that makes the responder indifferent between accepting and rejecting, and the offer is accepted immediately. Hence the firm offers w_f^*, the worker offers w_w^*, and*

$$r_w^*(w) = \mathbf{1}\{w \geq w_f^*\}, \qquad r_f^*(w) = \mathbf{1}\{w \leq w_w^*\}. \tag{30}$$

Proof. Under the Markov restriction, the continuation value following rejection does not depend on the rejected wage. The worker's value of agreement is strictly increasing in w, while the firm's is strictly decreasing. The optimal response rules in (6) are therefore cutoff rules. A proposer prefers the most favourable acceptable wage. An unacceptable offer delays agreement and exposes the negotiation to possible dissolution. The argument of Rubinstein (1982) and Binmore et al. (1986) applies directly, yielding immediate agreement at the responder's cutoff and acceptance at indifference. ■

Corollary 2 (Proposer advantage) *For either agent i, the value of agreement when proposing exceeds the value when responding:*

$$V_i(w_i^*) > V_i(w_j^*), \qquad j \neq i. \tag{31}$$

Consequently,

$$w_f^* < w_w^*. \tag{32}$$

Proof.

Note first that $V_i^0 \geq \frac{g_i}{1-\delta}$, since an agent in the search market can attend every negotiation and reject every offer, securing g_i in each period.

Suppose, towards a contradiction, that $V_i(w_j^*) \geq V_i(w_i^*)$. Then $V_i(w_j^*) \geq V_i(w_i^*) > V_i^0$, so $V_i(w_j^*)$ is the largest value agent i can hold at the start of any period. By Lemma 1, the responder is indifferent between accepting w_j^* and rejecting it:

$$V_i(w_j^*) = V_i^{\text{rej}}(R) \leq g_i + \delta V_i(w_j^*), \tag{33}$$

where the inequality holds because, after rejection, the agent receives g_i and then holds one of $V_i(w_i^*)$, $V_i(w_j^*)$ or V_i^0 in the following period, each of which is at most $V_i(w_j^*)$. Rearranging,

$$V_i(w_j^*) \leq \frac{g_i}{1-\delta} \leq V_i^0, \tag{34}$$

which contradicts $V_i(w_j^*) > V_i^0$. Hence $V_w(w_w^*) > V_w(w_f^*)$ and $V_f(w_f^*) > V_f(w_w^*)$. Since the worker's value is increasing and the firm's value decreasing in the wage, both imply $w_f^* < w_w^*$. ■

Lemma 2 (A single negotiation) *An agent with exactly one available negotiation attends it.*

Proof. Attendance has no direct cost and allows the agent to reject an unacceptable offer. Attending therefore weakly dominates attending neither, and is strictly preferable if the partner attends with positive probability. ■

Corollary 3 (New pairings) *Along the equilibrium path, every new pairing proceeds to bargaining and reaches immediate agreement. An agent expects any newly paired partner to attend with probability one.*

Proof. By Lemma 1, all offers are accepted along the equilibrium path, so no ongoing negotiations remain. A newly paired agent therefore has only one available negotiation and attends by Lemma 2. ■

Lemma 3 (Preference for a new negotiation) *An agent with both an ongoing and a new negotiation attends the new one in states (R, P), (P, P), and (R, R).*

Proof. A new partner attends with certainty by Corollary 3.

In state (R, P), the agent is proposer in the new negotiation and responder in the ongoing one. The new negotiation yields $V_i(w_i^*)$ with certainty, whereas the ongoing one yields at most $V_i(w_j^*)$. By Corollary 2, the new negotiation is strictly preferred.

In state (P,P), both negotiations yield $V_i(w_i^*)$ if attended by the partner. The ongoing partner is a responder and, by the (R,P) case, leaves whenever newly paired as proposer, an event of probability $\frac{\lambda_j}{2} > 0$. The new partner attends with certainty, so the new negotiation is strictly preferred.

In state (R,R), both negotiations yield $V_i(w_j^*)$ if attended by the partner. The ongoing partner is now a proposer and, by the (P,P) case, leaves whenever newly paired as proposer, again an event of probability $\frac{\lambda_j}{2} > 0$. The new negotiation is therefore strictly preferred.

Combining the (R,P) and (R,R) cases, a responder in an ongoing negotiation leaves upon any new pairing, an event of probability λ_j. This is the event used in Lemma 4. ■

The remaining state is (P,R). The ongoing negotiation gives the agent the proposer role but may not be attended by the partner; the new negotiation gives the responder role and takes place with certainty.

Lemma 4 (Preference for an ongoing negotiation) *An agent in state (P,R) attends the ongoing negotiation.*

Proof. Under the candidate strategies, a responder in an ongoing negotiation leaves upon any new pairing by Lemma 3. Thus an agent i who attends the ongoing negotiation as proposer obtains the expected value

$$(1-\lambda_j)V_i(w_i^*) + \lambda_j V_i^0,$$

where $j \neq i$ is the partner's type. Attending the new negotiation instead gives $V_i(w_j^*)$. We show that the former is larger.

Let $g_w = b$ and $g_f = -c$ denote the period payoffs of remaining in the search market. Under the candidate strategies, the value without an ongoing negotiation satisfies

$$V_i^0 = g_i + \delta\left[(1-\lambda_i)V_i^0 + \frac{\lambda_i}{2}V_i(w_i^*) + \frac{\lambda_i}{2}V_i(w_j^*)\right]. \tag{35}$$

The responder is indifferent between accepting w_j^* and rejecting it. After rejection they become proposer in the ongoing negotiation. They attend a new negotiation if paired as proposer, an event of probability $\lambda_i/2$; otherwise, they choose the ongoing negotiation, which continues if partner j has not been newly paired. Thus

$$\begin{aligned} V_i(w_j^*) = g_i + \delta\Bigg[&\frac{\lambda_i}{2}V_i(w_i^*) \\ &+ \left(1-\frac{\lambda_i}{2}\right)\left((1-\lambda_j)V_i(w_i^*) + \lambda_j V_i^0\right)\Bigg]. \end{aligned} \tag{36}$$

Subtracting (35) from (36) gives

$$\left(1+\frac{\delta\lambda_i}{2}\right)\left[V_i(w_j^*)-V_i^0\right]=\delta(1-\lambda_j)\left(1-\frac{\lambda_i}{2}\right)\left[V_i(w_i^*)-V_i^0\right]. \tag{37}$$

Because $\delta<1$, $\lambda_i,\lambda_j\in(0,1)$ and $V_i(w_i^*)>V_i^0$, this implies

$$V_i(w_j^*)-V_i^0<(1-\lambda_j)\left[V_i(w_i^*)-V_i^0\right]. \tag{38}$$

Therefore

$$V_i(w_j^*)<(1-\lambda_j)V_i(w_i^*)+\lambda_jV_i^0,$$

so the ongoing negotiation is strictly preferred in state (P,R). ■

A.1.2 Existence and uniqueness

Existence. We establish existence by constructing the equilibrium values and wage offers.

Since wages are transfers, the joint value of employment is independent of the wage. Define the total surplus relative to continued search as

$$S=V_w(w)+V_f(w)-V_w^0-V_f^0. \tag{39}$$

Adding the employment value equations gives

$$[1-\delta(1-s)]S=p-(1-\delta)(V_w^0+V_f^0). \tag{40}$$

We first express the equilibrium bargaining values in terms of S. Evaluating (39) at $w=w_w^*$ and at $w=w_f^*$ gives

$$S=\left[V_w(w_w^*)-V_w^0\right]+\left[V_f(w_w^*)-V_f^0\right]=\left[V_w(w_f^*)-V_w^0\right]+\left[V_f(w_f^*)-V_f^0\right].$$

Together with the indifference conditions (37) for workers and for firms, this is a linear system in the four differences $V_i(w_k^*)-V_i^0$, $i,k\in\{w,f\}$, whose unique solution is, for $i\in\{w,f\}$ and $j\neq i$,

$$\begin{aligned}V_i(w_i^*)-V_i^0&=\frac{2+\delta\lambda_i}{2+\delta[1+(1-\lambda_w)(1-\lambda_f)]}S,\\ V_i(w_j^*)-V_i^0&=\frac{\delta(1-\lambda_j)(2-\lambda_i)}{2+\delta[1+(1-\lambda_w)(1-\lambda_f)]}S.\end{aligned} \tag{41}$$

Next, substituting (41) into the search value equation (35) and rearranging yields

$$V_i^0 = \frac{g_i}{1-\delta} + \frac{\delta\lambda_i[2+\delta(2-2\lambda_j+\lambda_i\lambda_j)]}{2(1-\delta)[2+\delta(1+(1-\lambda_w)(1-\lambda_f))]}S. \tag{42}$$

Thus, the outside values are uniquely determined for any given S.

Finally, adding (42) for workers and firms and substituting into (40) gives

$$S = \frac{p-b+c}{1-\delta(1-s)+\frac{\delta}{2}\sum_{i\in\{w,f\}}\lambda_i\frac{2+\delta(2-2\lambda_j+\lambda_i\lambda_j)}{2+\delta[1+(1-\lambda_w)(1-\lambda_f)]}}. \tag{43}$$

The denominator is strictly positive since $\delta \in (0,1)$, $s \in (0,1)$, and $\lambda_w, \lambda_f \in (0,1)$. Under the assumption that there are gains from agreement such that $p-b+c>0$, equation (43) therefore determines a unique $S>0$.

Given S, equation (42) uniquely determines V_w^0 and V_f^0, while (41) determines the employment values associated with each bargaining role. In particular, $S>0$ implies $V_i(w_i^*) > V_i^0$ and $V_i(w_j^*) > V_i^0$ for both types, which is the condition used in Corollary 2, Lemma 4 and the uniqueness argument below; and since $2+\delta\lambda_i > \delta(1-\lambda_j)(2-\lambda_i)$, (41) delivers (31) directly.

Finally, since $V_w(w)$ is strictly increasing in w, the equilibrium wage offers are uniquely determined from (1):

$$w_i^* = [1-\delta(1-s)]V_w(w_i^*) - \delta s V_w^0, \qquad i \in \{w,f\}. \tag{44}$$

We have thus established the existence of values S, V_w^0, V_f^0, and wages w_f^* and w_w^* consistent with the equilibrium conditions. Together with the strategies characterised in the preceding lemmas, these constitute a Markov perfect equilibrium.

Uniqueness. Given stationary attendance strategies, the continuation value following rejection is independent of the wage offered. The uniqueness of the bargaining strategies therefore follows from the alternating offers argument of Rubinstein (1982) and Binmore et al. (1986).

We next establish uniqueness of attendance strategies. Under the convention that agents attend their only available negotiation, Lemma 2 determines behaviour in states with a single negotiation. By Lemma 3, attending the new negotiation is strictly optimal in states (R,P), (P,P), and (R,R). The only remaining state is (P,R), where the agent is the proposer in the ongoing negotiation and the responder in the new one.

Lemmas 1 to 3 do not depend on behaviour in state (P,R): the (R,P) case uses only

Corollaries 2 and 3, the (P,P) case uses the partner's behaviour in (R,P), and the (R,R) case uses the partner's behaviour in (P,P). Behaviour in (P,R) can therefore be determined one type at a time, holding the other type's behaviour in (P,R) arbitrary.

Suppose, towards a contradiction, that an agent of type i attends the new negotiation in state (P,R) with probability $\pi \in (0,1]$. After rejecting an offer, agent i is proposer in the ongoing negotiation. With probability $\lambda_i/2$ they are newly paired as proposer and attend the new negotiation; with probability $\lambda_i/2$ they are newly paired as responder and attend the new negotiation with probability π; otherwise they attend the ongoing negotiation, which continues if partner j has not been newly paired. The responder's indifference condition therefore becomes

$$\begin{aligned} V_i(w_j^*) = g_i + \delta \Bigg[& \frac{\lambda_i}{2} V_i(w_i^*) + \frac{\lambda_i}{2} \Big(\pi V_i(w_j^*) + (1-\pi)\big((1-\lambda_j)V_i(w_i^*) + \lambda_j V_i^0\big) \Big) \\ & + (1-\lambda_i)\big((1-\lambda_j)V_i(w_i^*) + \lambda_j V_i^0\big) \Bigg], \end{aligned} \tag{45}$$

where $j \neq i$. Subtracting the search value equation (35), we obtain

$$\begin{aligned} V_i(w_j^*) - V_i^0 = {} & \delta(1-\lambda_i)(1-\lambda_j)\big[V_i(w_i^*) - V_i^0\big] \\ & + \delta \frac{\lambda_i}{2}(1-\pi)\Big((1-\lambda_j)\big[V_i(w_i^*) - V_i^0\big] - \big[V_i(w_j^*) - V_i^0\big]\Big). \end{aligned} \tag{46}$$

Setting $\pi = 0$ recovers (37).

Attending the new negotiation with positive probability in state (P,R) is optimal only if

$$V_i(w_j^*) - V_i^0 \geq (1-\lambda_j)\big[V_i(w_i^*) - V_i^0\big]. \tag{47}$$

Under (47), the last term in (46) is non-positive, so

$$V_i(w_j^*) - V_i^0 \leq \delta(1-\lambda_i)(1-\lambda_j)\big[V_i(w_i^*) - V_i^0\big] < (1-\lambda_j)\big[V_i(w_i^*) - V_i^0\big], \tag{48}$$

where the strict inequality uses $\delta(1-\lambda_i) < 1$, $\lambda_j < 1$ and $V_i(w_i^*) > V_i^0$. This contradicts (47). Hence no strategy, pure or mixed, that attends the new negotiation in state (P,R) with positive probability can be part of an equilibrium.

Therefore, an agent must attend the ongoing negotiation in state (P,R). Together with the preceding lemmas, this uniquely determines the attendance strategies in (7). Given these strategies, the equilibrium value equations uniquely determine the wage offers, as established above. This completes the proof of Proposition 1.

A.2 Proof of Proposition 2

Differentiating the average bargaining power given by Equation (15) with respect to labour market tightness yields

$$\begin{aligned}\frac{\partial \bar{\beta}}{\partial \theta} = \frac{\delta}{2\left[2+\delta(1+(1-\lambda_w)(1-\lambda_f))\right]^2} \\ \times \Big[\lambda_w'(\theta)\left(2+\delta[1+(1-\lambda_f)^2]\right) \\ -\lambda_f'(\theta)\left(2+\delta[1+(1-\lambda_w)^2]\right)\Big].\end{aligned} \tag{49}$$

The denominator is strictly positive. Since $\lambda_w'(\theta) > 0$, $\lambda_f'(\theta) < 0$, and both expressions multiplying these derivatives are positive, the numerator is also strictly positive. Therefore,

$$\frac{\partial \bar{\beta}}{\partial \theta} > 0$$

for every $\delta \in (0,1)$. □

A.3 Equilibrium wage expressions

We derive the equilibrium wages after substituting the search values obtained in Appendix A.1.2.

From (42) and the equilibrium bargaining shares, the search values satisfy

$$\begin{aligned}(1-\delta)V_w^0 &= b + \delta\lambda_w\bar{\beta}S, \\ (1-\delta)V_f^0 &= -c + \delta\lambda_f(1-\bar{\beta})S.\end{aligned} \tag{50}$$

Substituting the worker's search value into (18), and using (40), gives

$$\begin{aligned}w_f^* &= b + \left[\delta\lambda_w\bar{\beta} + (1-\delta(1-s))\beta_f\right]S, \\ w_w^* &= b + \left[\delta\lambda_w\bar{\beta} + (1-\delta(1-s))\beta_w\right]S.\end{aligned} \tag{51}$$

The denominator of the total surplus expression (43) can equivalently be written as

$$1-\delta(1-s)+\delta\left[\lambda_w\bar{\beta}+\lambda_f(1-\bar{\beta})\right].$$

Substituting the equilibrium total surplus, we obtain

$$\begin{aligned} w_f^* &= b + \frac{\delta\lambda_w\bar{\beta} + [1-\delta(1-s)]\beta_f}{1-\delta(1-s)+\delta[\lambda_w\bar{\beta}+\lambda_f(1-\bar{\beta})]}(p-b+c), \\ w_w^* &= b + \frac{\delta\lambda_w\bar{\beta} + [1-\delta(1-s)]\beta_w}{1-\delta(1-s)+\delta[\lambda_w\bar{\beta}+\lambda_f(1-\bar{\beta})]}(p-b+c). \end{aligned} \tag{52}$$

Finally, taking the average across the two equally likely proposer roles gives

$$\bar{w} = b + \frac{\bar{\beta}[1-\delta(1-s)+\delta\lambda_w]}{1-\delta(1-s)+\delta[\lambda_w\bar{\beta}+\lambda_f(1-\bar{\beta})]}(p-b+c). \tag{53}$$

Together with the equilibrium bargaining shares in (14) and (15), these expressions characterise wages entirely in terms of the model's primitives and labour market tightness.

A.4 Asymmetric first mover probabilities

We extend the equilibrium surplus division to the case in which the firm makes the initial offer with probability $\mu \in [0,1]$ and the worker with probability $1-\mu$. The equilibrium attendance strategies remain unchanged, but the search values and the responder's continuation values depend on μ.

The search value equations become

$$\begin{aligned} V_w^0 &= b + \delta\Big[(1-\lambda_w)V_w^0 \\ &\qquad + \lambda_w\big(\mu V_w(w_f^*) + (1-\mu)V_w(w_w^*)\big)\Big], \\ V_f^0 &= -c + \delta\Big[(1-\lambda_f)V_f^0 \\ &\qquad + \lambda_f\big(\mu V_f(w_f^*) + (1-\mu)V_f(w_w^*)\big)\Big]. \end{aligned} \tag{54}$$

The worker's indifference condition when responding to the firm's offer is

$$\begin{aligned} V_w(w_f^*) = b + \delta\Bigg[&\lambda_w(1-\mu)V_w(w_w^*) \\ &+ \big(1-\lambda_w(1-\mu)\big)\Big((1-\lambda_f)V_w(w_w^*) + \lambda_f V_w^0\Big)\Bigg]. \end{aligned} \tag{55}$$

Similarly, the firm's indifference condition when responding to the worker's offer is

$$V_f(w_w^*) = -c + \delta \Bigg[\lambda_f \mu V_f(w_f^*) + \big(1 - \lambda_f \mu\big) \Big((1 - \lambda_w) V_f(w_f^*) + \lambda_w V_f^0 \Big) \Bigg]. \tag{56}$$

Subtracting the respective search values (54) and rearranging gives

$$\begin{aligned} S_w(w_f^*) &= \delta \frac{(1 - \lambda_f)(1 - \lambda_w(1 - \mu))}{1 + \delta \lambda_w \mu} S_w(w_w^*), \\ S_f(w_w^*) &= \delta \frac{(1 - \lambda_w)(1 - \lambda_f \mu)}{1 + \delta \lambda_f (1 - \mu)} S_f(w_f^*). \end{aligned} \tag{57}$$

Since total surplus does not depend on the wage, these conditions determine the worker's equilibrium surplus shares for each proposer role. We obtain

$$\begin{aligned} \frac{\beta_f}{1 - \beta_f} = \delta (1 - \lambda_f) & \frac{1 - \lambda_w(1 - \mu)}{1 + \delta \lambda_f (1 - \mu)} \\ & \times \frac{1 - \delta[1 - \lambda_w - \lambda_f(1 - \mu \lambda_w)]}{1 - \delta[1 - \lambda_f - \lambda_w(1 - (1 - \mu)\lambda_f)]}. \end{aligned} \tag{58}$$

Similarly, when the worker makes the initial offer,

$$\begin{aligned} \frac{\beta_w}{1 - \beta_w} = \frac{1}{\delta(1 - \lambda_w)} & \frac{1 + \delta \lambda_w \mu}{1 - \lambda_f \mu} \\ & \times \frac{1 - \delta[1 - \lambda_w - \lambda_f(1 - \mu \lambda_w)]}{1 - \delta[1 - \lambda_f - \lambda_w(1 - (1 - \mu)\lambda_f)]}. \end{aligned} \tag{59}$$

The expressions capture the effects of partner availability, outside opportunities, and the relative costs of delaying agreement. For general μ, these forces need not have the same relative importance for workers and firms.

The worker's expected bargaining power is

$$\bar{\beta}(\theta; \mu) = \mu \beta_f(\theta; \mu) + (1 - \mu) \beta_w(\theta; \mu). \tag{60}$$

Setting $\mu = \frac{1}{2}$ recovers the equilibrium bargaining shares in (14) and (15).

A.5 Bargaining power and market tightness

In the symmetric benchmark, where firms and workers are equally likely to make the first offer, worker bargaining power is strictly increasing in labour market tightness for any pairing technology satisfying the assumptions of the model. When first mover probabilities are asymmetric, however, the relationship is not necessarily monotonic. Its sign depends on how the pairing probabilities of workers and firms respond to changes in market tightness.

To illustrate the role of these responses, consider the limiting case in which firms always make the initial offer, $\mu \to 1$, and the discount factor approaches one, $\delta \to 1$. We consider this case simply to make the algebra lighter as the aim is to show the intuition.

Since firms make all initial offers, expected worker bargaining power coincides with the share obtained when the firm proposes, $\bar{\beta} = \beta_f$. Taking the limit of the equilibrium surplus shares and using $\lambda_w = \theta\lambda_f$, we obtain

$$\lim_{\substack{\mu\to 1\\ \delta\to 1}} \bar{\beta}(\theta;\mu) = \frac{(1-\lambda_f)(1+\theta-\lambda_w)}{1+\theta+(1-\lambda_f)(1+\theta-\lambda_w)}. \tag{61}$$

The expression illustrates two opposing effects of tighter labour markets. A reduction in the firm's probability of finding another worker lowers the risk that the firm abandons the negotiation, strengthening the worker's bargaining position. At the same time, an increase in the worker's probability of finding another firm raises the opportunity cost of prolonging the negotiation. Whether worker bargaining power increases therefore depends on the relative responsiveness of these probabilities to tightness.

Pairing elasticities and bargaining power. To characterise this relationship, let $P(N_u, N_v)$ denote the aggregate number of pairings generated by unemployed workers and vacancies. The pairing probabilities are given by

$$\lambda_w(\theta) = \frac{P(N_u,N_v)}{N_u}, \qquad \lambda_f(\theta) = \frac{P(N_u,N_v)}{N_v}. \tag{62}$$

Define the elasticities of aggregate pairings with respect to unemployment and vacancies as

$$\eta_u = \frac{\partial \log P(N_u,N_v)}{\partial \log N_u}, \qquad \eta_v = \frac{\partial \log P(N_u,N_v)}{\partial \log N_v}. \tag{63}$$

Under constant returns to scale, $\eta_u + \eta_v = 1$. Since $\theta = N_v/N_u$, these elasticities determine the responsiveness of the pairing probabilities to market tightness:

$$\frac{\partial \log \lambda_w}{\partial \log \theta} = \eta_v, \qquad \frac{\partial \log \lambda_f}{\partial \log \theta} = -\eta_u. \tag{64}$$

Differentiating (61) with respect to market tightness and substituting (64) yields

$$\frac{d}{d\theta}\left[\lim_{\substack{\mu\to 1\\ \delta\to 1}} \bar{\beta}(\theta;\mu)\right] = \frac{\lambda_f(1-\lambda_f)}{\left[1+\theta+(1-\lambda_f)(1+\theta-\lambda_w)\right]^2} \times \left\{\eta_u\left[\theta+\frac{(1+\theta)(1+\theta-\lambda_w)}{\theta(1-\lambda_f)}\right]-\eta_v\right\}. \tag{65}$$

Since the multiplicative term outside the braces is strictly positive, worker bargaining power increases with tightness if and only if

$$\eta_u\left[\theta+\frac{(1+\theta)(1+\theta-\lambda_w)}{\theta(1-\lambda_f)}\right] > \eta_v. \tag{66}$$

The condition highlights the importance of relative pairing elasticities. A higher elasticity with respect to unemployment, η_u, implies that the firm's probability of finding another worker decreases more strongly as the market tightens. This reduces the risk of losing the current negotiation. Conversely, a higher elasticity with respect to vacancies, η_v, implies that the worker's probability of finding another firm increases more strongly, raising the opportunity cost of delay. The first effect must be sufficiently strong relative to the second for worker bargaining power to increase.

An instructive counterexample is a Leontief pairing technology,

$$P(N_u, N_v) = \rho \min\{N_u, N_v\}, \qquad 0 < \rho < 1. \tag{67}$$

For $\theta < 1$, the pairing probabilities are $\lambda_f = \rho$ and $\lambda_w = \rho\theta$. Consequently, $\eta_u = 0$ and $\eta_v = 1$. The firm's pairing probability is locally independent of tightness, whereas the worker's pairing probability increases with it. Condition (66) is therefore violated, and worker bargaining power decreases with tightness in this region.

This limiting case establishes that the positive relationship between bargaining power and tightness need not hold under every pairing technology when first mover probabilities are asymmetric. Whether it holds under empirically relevant conditions depends on the elasticities of the pairing function.

Cobb–Douglas and CES pairing functions. We consider two commonly used constant-returns-to-scale pairing technologies. The first is the Cobb–Douglas function,

$$P(N_u, N_v) = \rho N_u^{\alpha} N_v^{1-\alpha}, \tag{68}$$

where $\rho > 0$ denotes pairing efficiency and $\alpha \in (0,1)$ is the elasticity with respect to unemployment. It implies

$$\lambda_w(\theta) = \rho\theta^{1-\alpha}, \qquad \lambda_f(\theta) = \rho\theta^{-\alpha}, \tag{69}$$

with constant elasticities $\eta_u = \alpha$ and $\eta_v = 1 - \alpha$.

The second is the CES pairing function,

$$P(N_u, N_v) = \rho\left[aN_u^{\frac{\sigma-1}{\sigma}} + (1-a)N_v^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}}, \tag{70}$$

where $a \in (0,1)$ and $\sigma > 0$ denotes the elasticity of substitution between unemployed workers and vacancies. For $\sigma \neq 1$, the implied pairing probabilities are

$$\lambda_w(\theta) = \rho\left[a + (1-a)\theta^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}}, \qquad \lambda_f(\theta) = \frac{\lambda_w(\theta)}{\theta}. \tag{71}$$

Unlike Cobb–Douglas, the CES specification allows the pairing elasticities to vary with tightness:

$$\eta_u(\theta) = \frac{a}{a + (1-a)\theta^{\frac{\sigma-1}{\sigma}}}, \qquad \eta_v(\theta) = \frac{(1-a)\theta^{\frac{\sigma-1}{\sigma}}}{a + (1-a)\theta^{\frac{\sigma-1}{\sigma}}}. \tag{72}$$

The Cobb–Douglas function is obtained as the limit when $\sigma \to 1$, with $\alpha = a$.

Both specifications allow us to evaluate the role of matching elasticities in determining bargaining power. We next use empirical estimates from the matching-function literature to select parameter values and examine whether the positive relationship between tightness and bargaining power persists when first mover probabilities are asymmetric.

Empirical evidence on matching elasticities The early empirical literature is surveyed by Petrongolo and Pissarides (2001), who document broad support for CRS Cobb–Douglas matching functions.

To choose empirically relevant parameters, I draw on the comparison of U.S. estimates compiled by Bernstein et al. (2025), together with the studies reported therein. I focus on estimates obtained under CRS and retain both Cobb–Douglas and CES specifications.

Table 5 reports the elasticity with respect to vacancies and its CRS-implied counterpart with respect to unemployment. For CES, it also reports the estimated substitution elasticity σ in the functional-form column.

The Cobb–Douglas estimates span a relatively wide range, while the CES estimates generally place the elasticity of substitution close to one, although these estimates are imprecise. This suggests that Cobb–Douglas is a useful benchmark, while motivating the consideration of CES specifications that allow matching elasticities to vary with tightness.

Table 5: Empirical estimates of matching elasticities

Study	Functional form	η_v	η_u
Cobb–Douglas			
Blanchard and Diamond (1989)	Cobb–Douglas	0.54	0.46
Bleakley and Fuhrer (1997)	Cobb–Douglas	0.31–0.35	0.65–0.69
Shimer (2005)	Cobb–Douglas	0.28	0.72
Hall (2005)	Cobb–Douglas	0.77	0.23
Rogerson and Shimer (2011)	Cobb–Douglas	0.42	0.58
Michaillat and Saez (2021)	Cobb–Douglas	0.51–0.61	0.39–0.49
Borowczyk-Martins et al. (2013)	Cobb–Douglas	0.70	0.30
Sahin et al. (2014)	Cobb–Douglas	0.24–0.66	0.34–0.76
Barnichon and Figura (2015)	Cobb–Douglas	0.34	0.66
Hall and Schulhofer-Wohl (2018)	Cobb–Douglas	0.35	0.65
CES			
Blanchard and Diamond (1989)	CES ($\sigma = 0.74$)	0.54	0.46
Shimer (2005)	CES ($\sigma = 1.06$)	0.28	0.72
Sahin et al. (2014)	CES ($\sigma = 0.9$–1.2)	0.24–0.66	0.34–0.76

Notes: η_v and η_u denote the elasticities of matches with respect to vacancies and unemployment, respectively. The unemployment elasticities are calculated as $1 - \eta_v$ under constant returns to scale, rather than independently estimated. For CES specifications, σ denotes the elasticity of substitution between unemployment and vacancies, while the matching elasticities refer to the reported reference estimates and need not remain constant across tightness.
Source: Estimates compiled by Bernstein et al. (2025) and the original studies listed in the table.

Numerical analysis under alternative pairing technologies. We now examine how equilibrium worker bargaining power responds to labour market tightness under different pairing technologies and empirically relevant matching elasticities. I use the equilibrium bargaining shares for general first mover probabilities, derived in (58) and (59), to calculate

$$\bar{\beta}(\theta;\mu) = \mu\beta_f(\theta;\mu) + (1-\mu)\beta_w(\theta;\mu). \tag{73}$$

For each pairing technology, I substitute the implied pairing probabilities $\lambda_w(\theta)$ and $\lambda_f(\theta)$ into these expressions and evaluate worker bargaining power over $\theta \in [0.1, 1.5]$.

Throughout the numerical analysis, I consider three probabilities that the firm makes the initial offer, $\mu \in \{0.1, 0.5, 0.9\}$. I set the discount factor to $\delta = 0.99$ and pairing efficiency to $\rho = 0.15$, holding both constant across specifications. The choice of ρ ensures that pairing probabilities remain strictly between zero and one over the range of tightness considered.

I first consider the Cobb–Douglas pairing function in (68). Guided by the empirical estimates in Table 5, I evaluate three unemployment elasticities, $\eta_u \in \{0.23, 0.5, 0.76\}$. The first and last correspond to the extremes of the estimates reported in the table, while the intermediate value represents a symmetric specification with $\eta_u = \eta_v = 0.5$. Figure 4 presents the results for the two extreme elasticities.

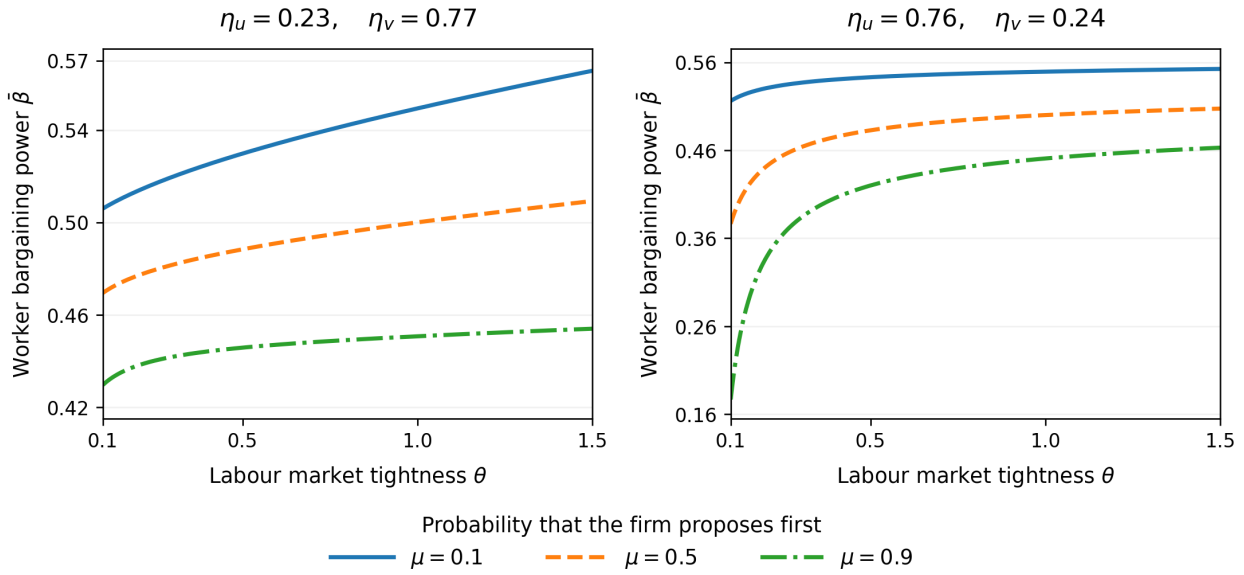


Figure 4: Expected worker bargaining power under Cobb–Douglas pairing. Unemployment elasticity: $\eta_u = 0.23$ (left) and $\eta_u = 0.76$ (right). Panel-specific vertical scales.

I then consider the CES pairing function in (70), allowing the matching elasticities to vary with tightness. Motivated by the empirical estimates in Table 5, I evaluate three elasticities of substitution, $\sigma \in \{0.74, 1.06, 1.2\}$. I set $a = 0.5$, so that the elasticities with respect to

unemployment and vacancies are both equal to 0.5 at $\theta = 1$. All other parameters remain unchanged. Figure 5 presents the results for the two extreme substitution elasticities.

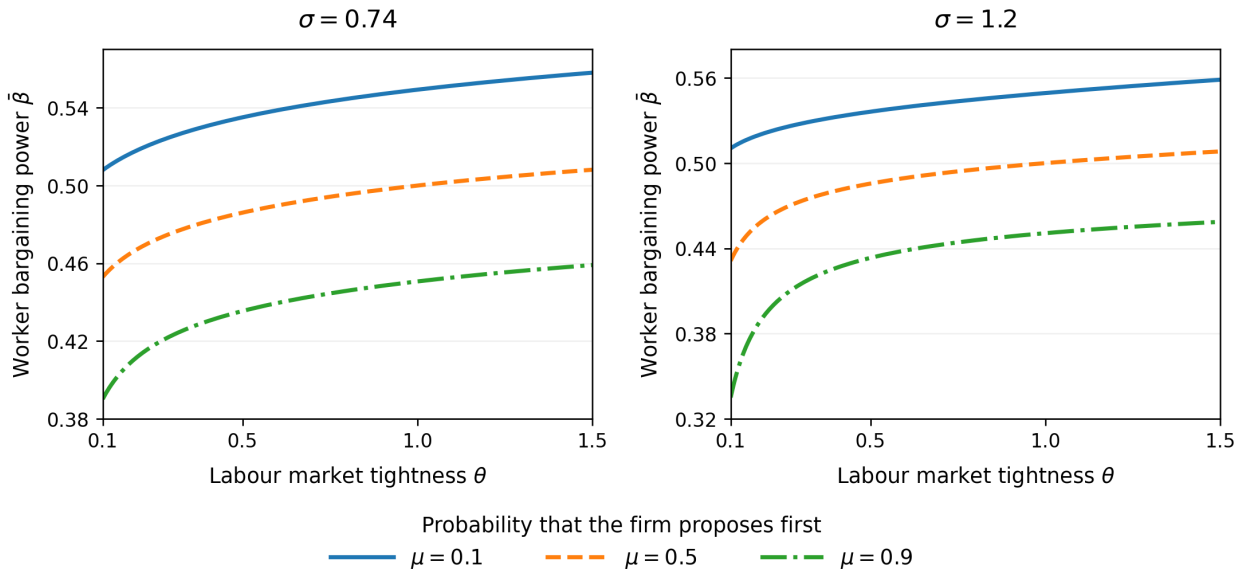


Figure 5: Expected worker bargaining power under CES pairing. Elasticity of substitution: $\sigma = 0.74$ (left) and $\sigma = 1.2$ (right). Panel-specific vertical scales.

A.6 Firm productivity heterogeneity

This appendix characterises equilibrium bargaining with heterogeneous firm productivity and proves the productivity results used in Section 3. As in the benchmark, I set $\mu = \frac{1}{2}$ and take labour market tightness θ as given. Firm productivity is permanent, with distribution F, and the productivity of a newly encountered firm is drawn independently from this distribution. Pairing probabilities do not depend on productivity. Throughout, I assume that employment generates positive total surplus $S(p) > 0$, with $S'(p) > 0$.

The employment value of a worker remains $V_w(w)$, while the firm's value of employment and of maintaining its vacancy now depend on its productivity: $V_f(w;p)$ and $V_f^0(p)$. The value of unemployment V_w^0 is common to all workers. Extending (8), define

$$S_w(w) = V_w(w) - V_w^0, \qquad S_f(w;p) = V_f(w;p) - V_f^0(p). \tag{74}$$

Thus, at either equilibrium wage offer,

$$\begin{aligned} S(p) &= S_w(w_f^*(p)) + S_f(w_f^*(p);p) \\ &= S_w(w_w^*(p)) + S_f(w_w^*(p);p). \end{aligned} \tag{75}$$

Search values and optimal bargaining. The optimality conditions in (4)–(6) continue to apply, with firm productivity added to the payoff-relevant state. In an equilibrium with immediate agreement, the values of searching satisfy

$$\begin{aligned} V_w^0 &= b + \delta \left[(1-\lambda_w) V_w^0 + \frac{\lambda_w}{2} \int \left(V_w(w_f^*(p')) + V_w(w_w^*(p')) \right) dF(p') \right], \\ V_f^0(p) &= -c + \delta \left[(1-\lambda_f) V_f^0(p) + \frac{\lambda_f}{2} \left(V_f(w_f^*(p); p) + V_f(w_w^*(p); p) \right) \right]. \end{aligned} \tag{76}$$

The firm's attendance decisions retain the role dependent structure in (7): a new worker does not change the firm's productivity or its payoffs conditional on the bargaining role. The worker, by contrast, compares both the bargaining role and the productivity of the new employer.

Consider first a worker who rejects the firm's offer $w_f^*(p)$. In the following period, the worker is the proposer in the ongoing negotiation. The incumbent firm remains available with probability $1-\lambda_f$. When a new pairing with a firm of productivity p' arrives, the worker can instead receive $w_w^*(p')$ if selected to propose, or $w_f^*(p')$ if selected to respond. Optimal attendance therefore yields

$$\begin{aligned} V_w(w_f^*(p)) = b + \delta \Bigg[& (1-\lambda_w) \left((1-\lambda_f) V_w(w_w^*(p)) + \lambda_f V_w^0 \right) \\ & + \frac{\lambda_w}{2} \sum_{j \in \{f,w\}} \int \max\left\{ (1-\lambda_f) V_w(w_w^*(p)) + \lambda_f V_w^0,\, V_w(w_j^*(p')) \right\} dF(p') \Bigg]. \end{aligned} \tag{77}$$

Subtracting the worker's search value in (76) and using (74) gives the surplus form

$$\begin{aligned} S_w(w_f^*(p)) = \delta \Bigg[& (1-\lambda_f) S_w(w_w^*(p)) \\ & - \frac{\lambda_w}{2} \sum_{j \in \{f,w\}} \int \min\left\{ (1-\lambda_f) S_w(w_w^*(p)), S_w(w_j^*(p')) \right\} dF(p') \Bigg]. \end{aligned} \tag{78}$$

The minimum terms reflect the difference between the opportunities available to an unemployed worker and those that improve on the expected value of continuing the incumbent negotiation.

Now consider a firm of productivity p that rejects the worker's offer $w_w^*(p)$. The firm becomes the proposer in the ongoing negotiation. It is paired with another worker with probability λ_f: when it can propose to that worker, it chooses the new negotiation; otherwise,

it continues with the incumbent worker if that worker also attends.

Let $\pi_R(p)$ be the probability that the incumbent worker abandons the negotiation following rejection, when the worker is the responder in the ongoing negotiation. The firm's probability of continuing with the incumbent worker, conditional on choosing the old negotiation, is $1 - \pi_R(p)$. Optimal worker attendance implies

$$\pi_R(p) = \frac{\lambda_w}{2} \sum_{j \in \{f,w\}} \int \mathbf{1}\left\{ S_w(w_j^*(p')) > \left(1 - \frac{\lambda_f}{2}\right) S_w(w_f^*(p)) \right\} dF(p'). \tag{79}$$

Here the factor $1 - \frac{\lambda_f}{2}$ is the probability that the incumbent firm attends when it is the proposer in the ongoing negotiation.

The firm's indifference condition is

$$\begin{aligned} V_f(w_w^*(p); p) = -c + \delta \Bigg[& \frac{\lambda_f}{2} V_f(w_f^*(p); p) \\ & + \left(1 - \frac{\lambda_f}{2}\right) \left((1 - \pi_R(p)) V_f(w_f^*(p); p) + \pi_R(p) V_f^0(p) \right) \Bigg]. \end{aligned} \tag{80}$$

Subtracting the firm's search value and rearranging yields

$$\left(1 + \frac{\delta \lambda_f}{2}\right) S_f(w_w^*(p); p) = \delta \left(1 - \frac{\lambda_f}{2}\right) (1 - \pi_R(p)) S_f(w_f^*(p); p). \tag{81}$$

These equations also verify the firm's proposed attendance rule. With a single negotiation, the firm attends it. When two negotiations are available, the firm weakly prefers the new one if its role is (P, P) or (R, R), since its conditional bargaining payoff is the same and the new partner attends with certainty; it prefers the new one in (R, P) because it becomes the proposer. Finally, in (P, R), (81) implies

$$S_f(w_w^*(p); p) < (1 - \pi_R(p)) S_f(w_f^*(p); p),$$

so the ongoing negotiation is strictly preferred. Thus the firm's attendance strategy coincides with (7).

Equations (76), (75), (78), and (81), together with optimal attendance, characterise equilibrium wages and surplus division across productivity types. We next establish monotonicity of the wage schedules, which justifies representing workers' attendance choices by productivity cutoffs.

Productivity and equilibrium wages.

Lemma 5 (Firm productivity and wages) *Conditional on the identity of the proposer, equilibrium wages are strictly increasing in firm productivity. For any $p' > p$,*

$$w_f^*(p') > w_f^*(p), \qquad w_w^*(p') > w_w^*(p). \tag{82}$$

Consequently, their equally weighted average is strictly increasing in firm productivity.

Proof. Fix $p' > p$, so that $S(p') > S(p)$. Suppose, towards a contradiction, that $w_f^*(p') \leq w_f^*(p)$. Since $S_w(w)$ is strictly increasing in the wage, this implies

$$S_w(w_f^*(p')) \leq S_w(w_f^*(p)). \tag{83}$$

The right-hand side of (78) is strictly increasing in the worker's proposer surplus $S_w(w_w^*(p))$. Indeed, the minimum of a fixed number and its continuation-surplus argument is nondecreasing and one-Lipschitz in that argument. Consequently, if the worker's proposer surplus changes by a positive amount, the right-hand side of (78) increases by at least $\delta(1-\lambda_f)(1-\lambda_w)$ times that amount. It increases by no more than $\delta(1-\lambda_f)$ times that amount. Thus the responder and proposer surpluses move strictly in the same direction, and (83) requires

$$S_w(w_w^*(p')) \leq S_w(w_w^*(p)). \tag{84}$$

Furthermore, the upper bound on the change gives

$$\begin{aligned} &S_w(w_f^*(p)) - S_w(w_f^*(p')) \\ &\quad \leq \delta(1-\lambda_f)\left[S_w(w_w^*(p)) - S_w(w_w^*(p'))\right] \\ &\quad \leq S_w(w_w^*(p)) - S_w(w_w^*(p')). \end{aligned} \tag{85}$$

By (83), the expected payoff from staying with the incumbent is weakly lower when its productivity is p' than when it is p. Hence every alternative that induces the worker to leave the firm of productivity p also induces departure from the firm of productivity p'. Equation (79) therefore implies

$$\pi_R(p') \geq \pi_R(p). \tag{86}$$

The firm's indifference condition (81) then requires

$$\frac{S_f(w_w^*(p');p')}{S_f(w_f^*(p');p')} \leq \frac{S_f(w_w^*(p);p)}{S_f(w_f^*(p);p)}. \tag{87}$$

But the surplus identities in (75) imply

$$\begin{aligned} &S_f(w_w^*(p');p') - S_f(w_w^*(p);p) \\ &\quad = S(p') - S(p) + S_w(w_w^*(p)) - S_w(w_w^*(p')), \\ &S_f(w_f^*(p');p') - S_f(w_f^*(p);p) \\ &\quad = S(p') - S(p) + S_w(w_f^*(p)) - S_w(w_f^*(p')). \end{aligned} \tag{88}$$

Both increments are strictly positive, and (85) implies that the firm's responder surplus rises by at least as much as its proposer surplus. Since the proposer obtains the larger strictly positive surplus, $S_f(w_f^*(p);p) > S_f(w_w^*(p);p) > 0$, it follows that

$$\frac{S_f(w_w^*(p');p')}{S_f(w_f^*(p');p')} > \frac{S_f(w_w^*(p);p)}{S_f(w_f^*(p);p)},$$

contradicting (87). We conclude that $w_f^*(p') > w_f^*(p)$.

Finally, the strictly increasing relation between the worker's two surpluses established from (78) implies $S_w(w_w^*(p')) > S_w(w_w^*(p))$, and hence $w_w^*(p') > w_w^*(p)$. ■

Productivity cutoffs and attendance. By Lemma 5, the worker's payoff from a new negotiation is strictly increasing in the productivity p' of the newly encountered firm. It follows that optimal attendance can be expressed by a cutoff $p^*(p;s_w)$ for each bargaining-role state $s_w = (q_w^o, q_w^n) \in \{P,R\}^2$, as in Proposition 3.

If the worker is the proposer in the ongoing negotiation, the incumbent firm attends with probability $1-\lambda_f$. When the worker is the responder, that probability is $1-\frac{\lambda_f}{2}$. The four interior cutoffs are defined by

$$\begin{aligned} S_w(w_w^*(p^*(p;(P,P)))) &= (1-\lambda_f)S_w(w_w^*(p)), \\ S_w\big(w_f^*(p^*(p;(P,R)))\big) &= (1-\lambda_f)S_w(w_w^*(p)), \\ S_w(w_w^*(p^*(p;(R,P)))) &= \left(1-\frac{\lambda_f}{2}\right) S_w(w_f^*(p)), \\ S_w\big(w_f^*(p^*(p;(R,R)))\big) &= \left(1-\frac{\lambda_f}{2}\right) S_w(w_f^*(p)). \end{aligned} \tag{89}$$

For each state, the worker chooses the new negotiation if $p' > p^*(p;s_w)$ and the ongoing negotiation if $p' < p^*(p;s_w)$. At equality, either choice is optimal; we select the new negotiation. When the indifference point lies outside the support of F, the worker always chooses the same negotiation for that state.

For completeness, the probabilities of abandoning an ongoing negotiation conditional on

the worker's role there are

$$\begin{aligned}\pi_P(p) &= \frac{\lambda_w}{2}\left[1 - F(p^*(p;(P,P))) + 1 - F(p^*(p;(P,R)))\right],\\ \pi_R(p) &= \frac{\lambda_w}{2}\left[1 - F(p^*(p;(R,P))) + 1 - F(p^*(p;(R,R)))\right].\end{aligned} \tag{90}$$

The second expression agrees with (79). For a continuous productivity distribution, the treatment of exact indifference is immaterial.

Proof of Corollary 1. **Proof.** Fix a state $s_w \in \{P,R\}^2$ and two incumbent productivities $p' > p$. By Lemma 5, the worker's equilibrium surplus from the incumbent negotiation is strictly higher at p' than at p, whether the worker is proposer or responder. The probability that the incumbent firm attends, equal to $1-\lambda_f$ or $1-\frac{\lambda_f}{2}$, depends on the worker's role but not on productivity. The right-hand side of the corresponding cutoff equation in (89) is therefore strictly higher at p'. Its left-hand side is a strictly increasing function of the new firm's productivity. Consequently, for interior cutoffs,

$$p^*(p';s_w) > p^*(p;s_w).$$

This establishes the claimed monotonicity. If the wage schedules are differentiable with strictly positive derivatives at the relevant productivities, implicit differentiation of (89) also gives $\frac{\partial p^*(p;s_w)}{\partial p} > 0$. ■

Equilibrium division of surplus. Using the same definition as in the benchmark, the worker's conditional bargaining shares are

$$\beta_f(p,\theta) = \frac{S_w(w_f^*(p))}{S(p)}, \qquad \beta_w(p,\theta) = \frac{S_w(w_w^*(p))}{S(p)}, \tag{91}$$

and expected bargaining power is

$$\bar{\beta}(p,\theta) = \frac{1}{2}\left[\beta_f(p,\theta) + \beta_w(p,\theta)\right]. \tag{92}$$

In particular, (81) implies

$$1 - \beta_w(p,\theta) = \frac{\delta\left(1-\frac{\lambda_f}{2}\right)(1-\pi_R(p))}{1+\frac{\delta\lambda_f}{2}}\left[1-\beta_f(p,\theta)\right], \tag{93}$$

while (78), divided by $S(p)$, gives

$$\beta_f(p,\theta) = \delta\left[(1-\lambda_f)\beta_w(p,\theta) \quad -\frac{\lambda_w}{2}\sum_{j\in\{f,w\}}\int \min\left\{(1-\lambda_f)\beta_w(p,\theta), \frac{S(p')}{S(p)}\beta_j(p',\theta)\right\} dF(p')\right]. \tag{94}$$

These conditions, together with (89) and (90), determine the productivity-dependent division of surplus. Unlike the homogeneous benchmark, the attractiveness of outside firms depends on the incumbent firm's productivity as well as on the bargaining roles.

A.7 Productivity pass-through across additional sectors

Figure 6 extends the sector specific analysis to the ten broad sectors not displayed in Figure 3. The specifications, regional TFPR-decile definitions, evaluation points, and confidence-interval construction are identical to those used in the main text.

The additional sectors display considerable heterogeneity in both the level and shape of productivity pass-through. Several profiles are approximately flat or decline towards the upper deciles, but some estimates are imprecise due to the smaller number of observations in these sectors.

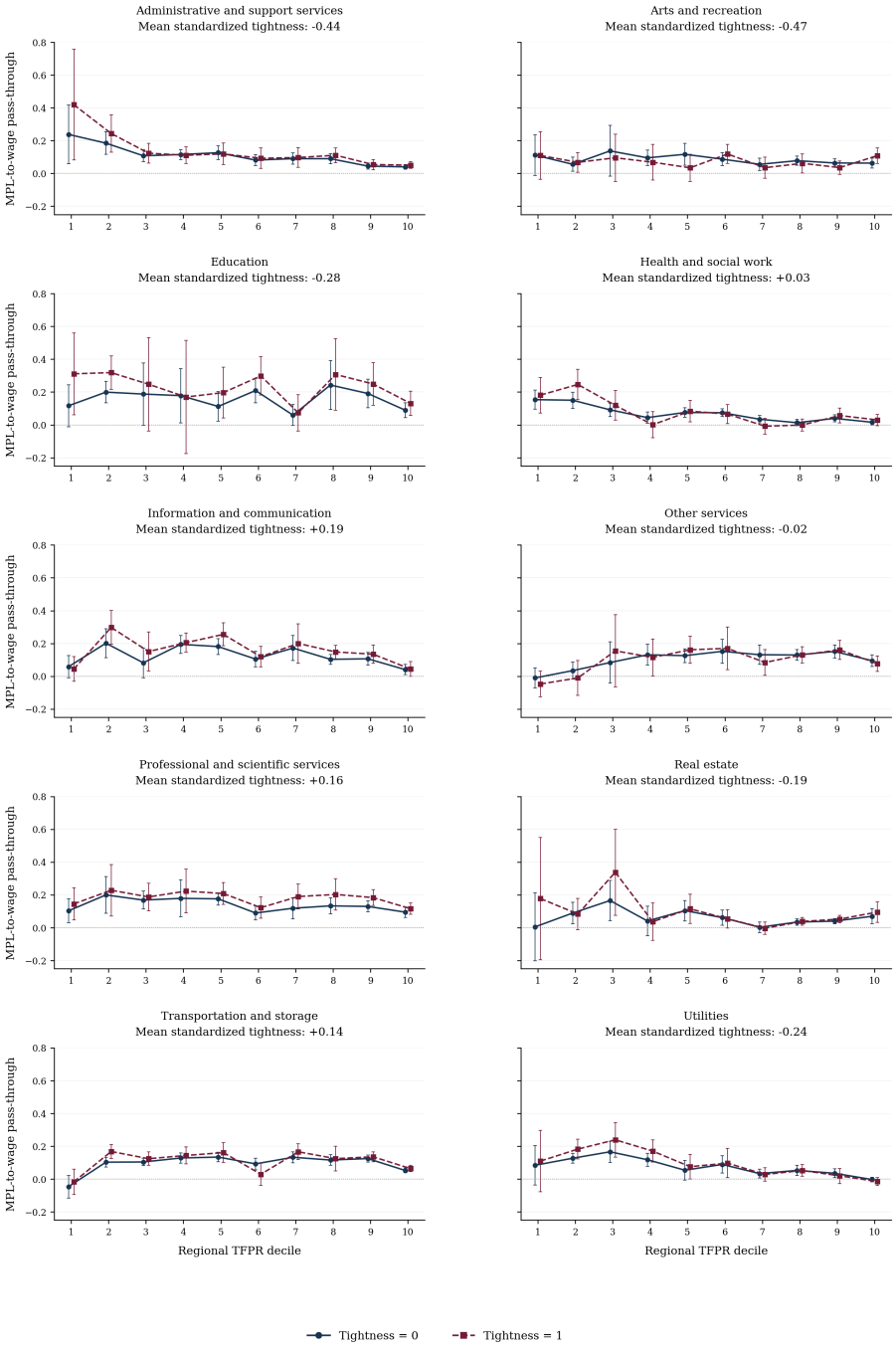


Figure 6: Productivity pass-through across firms: additional sectors

Notes: This figure reports the ten sectors not shown in Figure 3. Within each sector, the solid navy line shows implied MPL-to-wage pass-through at standardised labour market tightness equal to zero, and the dashed burgundy line shows pass-through at standardised tightness equal to one. Vertical bars indicate 95% confidence intervals. Panel headings report observation weighted sector mean standardised tightness.